

ACCORD SOFTWARE & SYSTEMS PVT LTD

TWSTFT NGS-T20 MODEM

USER MANUAL

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This user manual describes the installation and operation procedure of TWSTFT
MODEM

TABLE OF CONTENTS

1	introduction	4
1.1	Reference	4
1.2	Proprietary rights	4
1.3	Contact information	4
1.4	Organization of the manual	4
1.5	Acronyms and abbreviations.....	5
2	Two Way Signal Time & Frequency Transfer MODEM	6
2.1	OVERVIEW	6
2.2	TWSTFT MODEM Front Panel	7
2.3	TWSTFT MODEM Rear Panel.....	8
2.3.1.1	Spec	8
2.4	Specifications	9
2.4.1	Interface Specifications.....	9
2.4.2	Performance Specifications	10
2.4.3	Environmental Specifications	11
2.5	USE & CARE	12
3	setup guidelines	12
3.1	Antenna Setup.....	12
3.1.1	Antenna mounting considerations:	13
3.2	Power Input.....	14
3.2.1	Using AC Input Power	14
3.2.2	Using DC Input Power	15
3.3	MODEM Setup	15
3.3.1	Powering ON.....	15
3.3.2	Setting up IP Address.....	15
3.4	Accessing the WEB UI.....	17
4	Configuration & Operation Through LCD	18
4.1	TWSTFT	18
4.1.1	STATUS.....	18

4.1.2	CONFIGURATION	19
4.2	GNSS RX.....	20
4.2.1	CHANNEL DATA.....	20
4.2.2	GPS.....	20
4.2.3	IRNSS L5	20
4.2.4	IRNSS S	21
4.2.5	GLONASS	21
4.2.6	SBAS	21
4.2.7	CONFIGURATION	21
4.3	LAN PORTS	22
4.3.1	LAN-1 SETTINGS	22
4.3.2	LAN-2 SETTINGS	22
4.3.3	PTP	22
4.3.4	NTP.....	25
5	CALIBRATION.....	27
6	HARDWARE STATUS	27
7	Configuration & Operation Through WEB UI	27
5.1	Operating Web Ui	28
5.2	Twstft	29
5.3.1	Schedule File	31
5.3	Gnss Receiver	33
5.4	Lan1 And Lan2.....	35
5.5	Ptp	36
5.6	Ntp.....	37
5.7.1	To Modify NTP Service.....	38
5.7.2	To Modify NTP Message Authentication	38
5.7	Hardware Status And Configuration	39
5.8	Calibration.....	40
5.9	Maintenance And Support	40
8	Data Logging Through SD card	40
8.1	Folder creation Logic:	42

SPECIFICATIONS

9	Software Upgrade (SWUG)	43
10	Cable Details	44
10.1	DC-DC Input Power Cable.....	44
10.2	AC Input Power Cable	45
10.3	RS-422 Data Cable	45
10.3.1	NMEA Data OUT & 1PPS.....	45
10.3.2	10 MHz OUT	46
10.3.3	NMEA Data IN.....	46
11	Mechanical Specifications	47
11.1	Antenna.....	47
11.2	Antenna Mount.....	48
11.3	MODEM.....	49
	Appendix:	52
12	Troubleshooting	52
13	Frequently Asked Questions (FAQ)'s	53

1 INTRODUCTION

This manual describes how to set up, install and use the TWSTFT Modem.

1.1 REFERENCE

Following are the product references:

1.2 PROPRIETARY RIGHTS

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1.3 CONTACT INFORMATION

For assistance in setting up the TWSTFT MODEM or if any unforeseen problems arise, please contact technical support by either sending an email to supportigs@accord-soft.com or by submitting a query through the online portal www.accord-soft.com\csp-portal\.

For more information about Accord and its range of GNSS products, please visit www.accord-soft.com or email to contactus@accord-soft.com.

1.4 ORGANIZATION OF THE MANUAL

1.5 ACRONYMS AND ABBREVIATIONS

AC	Alternate Current
BNC	Bayonet Neill-Councilman
COM	Communication
CDROM	Compact Disc - Read Only Memory
C/N0	Carrier to Noise Ratio
DC	Direct Current
GAGAN	GPS Aided Geo Augmented Navigation
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GUI	Graphical User Interface
GLONA	Global Navigation Satellite System
IP	Internet Protocol
IRNSS	Indian Regional Navigation Satellite System
LCD	Liquid Crystal Display
LED	Light Emitting Diode
NMEA	National Marine Electronics Association
PC	Personal Computer
PVT	Position Velocity & Time
PRN	Pseudo Random Number
RF	Radio Frequency
RS-232	Recommended Standard No. 232
SBAS	Satellite Based Augmentation System
SMA	Sub-Miniature Version A
SWUG	Software Upgrade
TNC	Threaded Neill-Concelman
TOWC	Time Of Week Count
USB	Universal Serial Bus

2 TWO WAY SIGNAL TIME & FREQUENCY TRANSFER MODEM

Before looking into the working of TWSTFT NGS-T20 MODEM, let's peek into the overview, features, use and care of TWSTFT MODEM.

2.1 OVERVIEW

This section gives brief description of the steps need to be followed prior using TWSTFT NGS-T20 MODEM. This Includes:

- **Installation:** Hardware setup, mechanical installations & physical connections.
- **Setup:** Establish basic access to the unit, so as to allow Web user interface and communication through LCD.
- **Configuration:** Allow Web UI & LCD for configuring and accessing modes, protocols & other settings.

2.2 TWSTFT MODEM FRONT PANEL

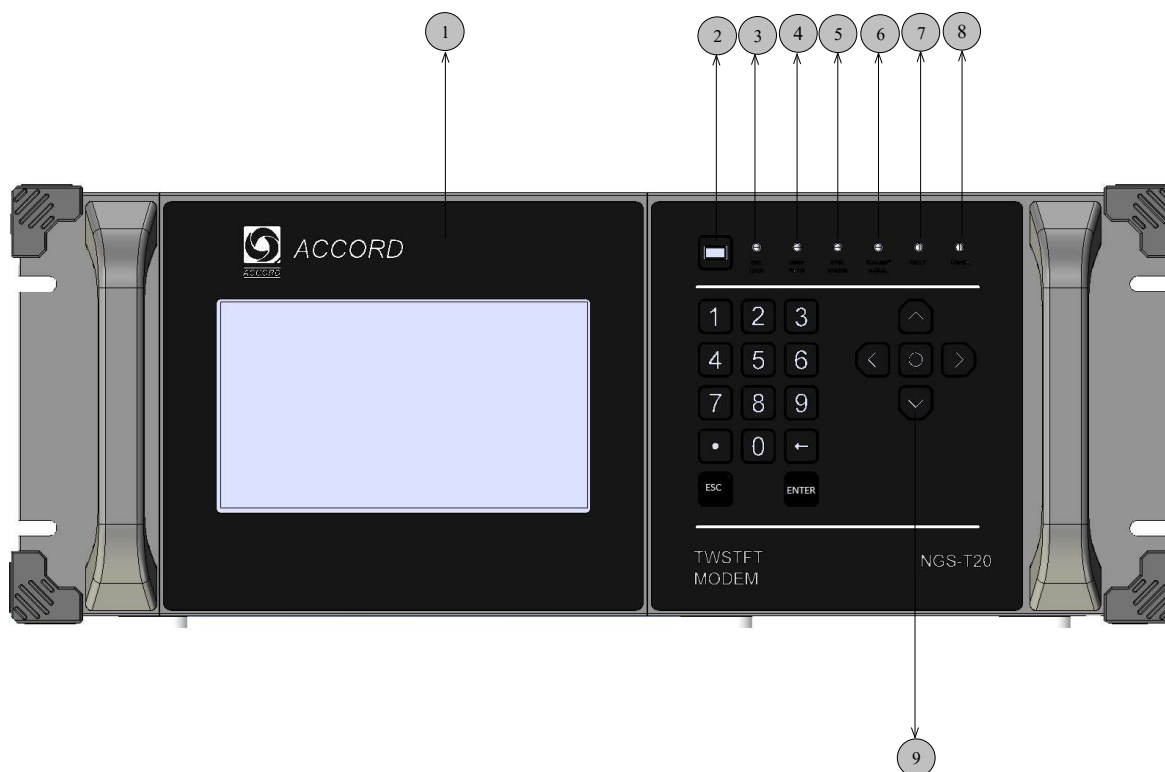


FIGURE 1: FRONT PANEL

REFERENCE	DESCRIPTION
1	LCD
2	DISPLAY ON/OFF
3	OSC LOCK
4	GNSS RX FIX
5	SYNC STATUS
6	TRANSMIT SIGNAL
7	FAULT
8	POWER
9	NAVIGATION KEYS

TABLE 1: FRONT PANEL SPECIFICATIONS

2.3 TWSTFT MODEM REAR PANEL

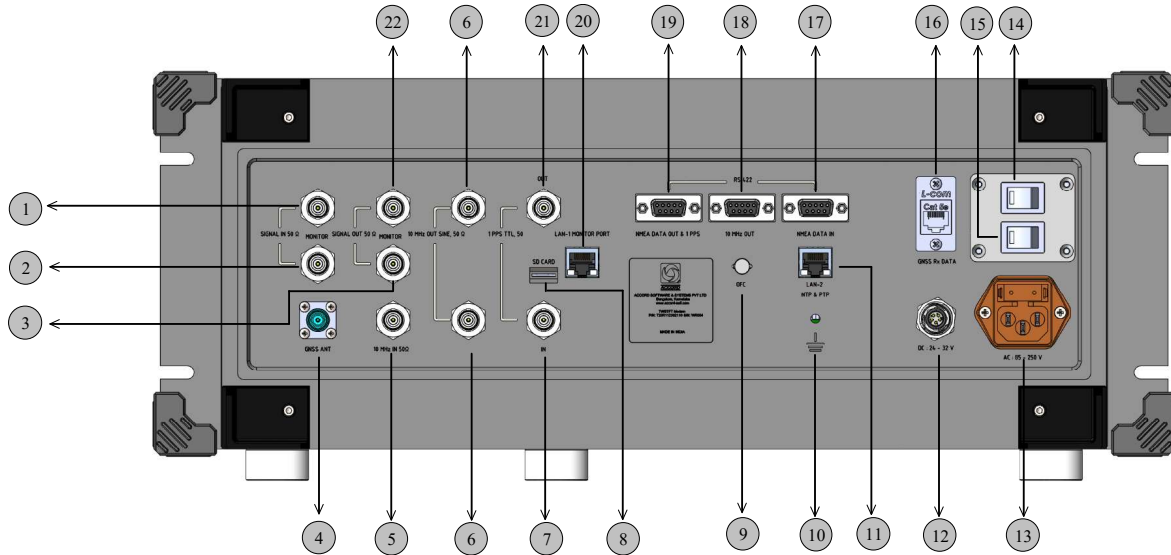


FIGURE 2: REAR PANEL

REFERENCE	TYPE	DESCRIPTION
1	BNC	IF SIGNAL IN MONITOR(50 ohm)
2	BNC	IF SIGNAL IN(50 ohm)
3	BNC	IF SIGNAL OUT(50 ohm)
4	TNC	GNSS ANTENNA IN
5	BNC	10 MHz IN(50 ohm)
6	BNC	10 MHz OUT (50 ohm)
7	BNC	1PPS TTL IN(50 ohm)
8	CONNECTOR	SD CARD CONNECTOR
9	CONNECTOR	OFC
10	STUD	CHASSIS GROUND
11	RJ-45	LAN-2 (NTP & PTP)
12	CONNECTOR	DC INPUT(24-32 V)
13	CONNECTOR	AC INPUT(85-250 V)
14	SWITCH	AC SWITCH
15	SWITCH	DC SWITCH
16	RJ-45(CAT6)	GNSS RECEIVER DATA IN
17	DB-9	NMEA DATA IN(RS-422)
18	DB-9	10 MHz OUT(RS-422)
19	DB-9	NMEA DATA OUT & 1 PPS
20	RJ-45	LAN-1 MONITOR PORT
21	BNC	1PPS TTL OUT(50 ohm)

TABLE 2: REAR PANEL SPECIFICATIONS

2.4 SPECIFICATIONS

2.4.1 INTERFACE SPECIFICATIONS

Parameter	Connector	Specifications
Power Input (DC)	4 pin circular SAMTEC, Part number: ACPK- 12-01-G-S-M	24-32 V DC 10.5 W typical power consumption 11.5 W max power consumption
Power Input (AC)		100-240 V AC, 50/60 Hz $\pm 10\%$
Antenna Input	TNC female	50 Ω nominal impedance Antenna power: 4.6 to 4.8 VDC, 200 mA Max
10 MHz input	BNC female	50 Ω nominal impedance Sine/Square wave input with power 0 dBm ± 3 dB
10 MHz Output	BNC female	50 Ω nominal impedance Sine wave output with power 0 dBm ± 3 dB
Signal IN	BNC female	50 Ω nominal impedance Sine wave input with power 0 dBm ± 3 dB
Signal OUT	BNC female	50 Ω nominal impedance Sine wave output with power 0 dBm ± 3 dB
NMEA Data IN	DB9 female	RS422 port Baud rate supported: 4800, 9600, 19200, 38400, 57600, 115200, 230400
NMEA Data OUT	DB9 female	RS422 port
10 MHz OUT	DB9 female	RS422 port
SD card	Micro SD card	SDHC, with class 4 to 10

SPECIFICATIONS

	connector	
1PPS IN	BNC female	TTL, 0- 5 VDC input
1PPS OUT	BNC female	TTL, 0 -5 VDC output
LAN-1 Monitor Port	RJ-45 jack	10/100 base T Ethernet support
LAN-2 NTP & PTP	RJ-45 jack	10/100 base T Ethernet support
OFC Data OUT	T-package female Part No: HFBR-1414MZ	Signal rate up to 160 MBd. Supports 820 nm wavelength technology
EARTH STUD	Part no: 753	Supports up to 15A
GNSS RX DATA	RJ-45 jack	100/1000 base T Ethernet support

2.4.2 PERFORMANCE SPECIFICATIONS

Parameter	Performance Specifications	
Accuracy	Time Synchronization between Master clock & Slave clock	<100 ns (1σ)
Precision	Time Synchronization between Master clock & Slave clock	Better than 100 ns
Holdover accuracy		≤ 5 microseconds over 24 hours at ambient temperature
Security	Encryption	Serpent-256 block cipher
	Authentication	SHA-2
	Host communication	TCP/IP

SPECIFICATIONS

Protocols	Timing Protocol	NTP v3 & v4
	Timing Protocol	PTP IEEE 1588
	Management Protocol	SNMP v1/v2c
User Interface	WEB UI	TCP/HTTP
IF Input Signal Specification	1.Programmable IF Center Frequency 2. Bandwidth 3. Power Level	140-160 MHz 4 MHz around center 0 dBm +/- 3dB at 50 ohm
IF Output Signal Specification	1.Programmable IF Center Frequency 2. Bandwidth 3. Power Level	140-160 MHz 4 MHz around center 0 dBm +/- 3dB at 50 ohm
Internal Clock Source	Rubidium Atomic standard clock source	
External Reference Input Signal	10 MHz	0 dBm, 50 ohm Sine Wave
	1 PPS	CMOS 3.3 V
Reference Output Signal	1 PPS	RS-422 Level

2.4.3 ENVIRONMENTAL SPECIFICATIONS

Parameter	Specifications	
Temperature	Operational	-10°C to +55°C
	Storage	-40°C to +85°C
Humidity		

Altitude		

2.5 USE & CARE

The TWSTFT MODEM is a high-precision electronic device and should be treated with reasonable care.

	Do not operate TWSTFT MODEM outside the specified temperature range. It can degrade the performance or damage permanently.
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3 SETUP GUIDELINES

3.1 ANTENNA SETUP

Figure 3 highlights the procedure for mounting antenna using the supplied antenna mount.

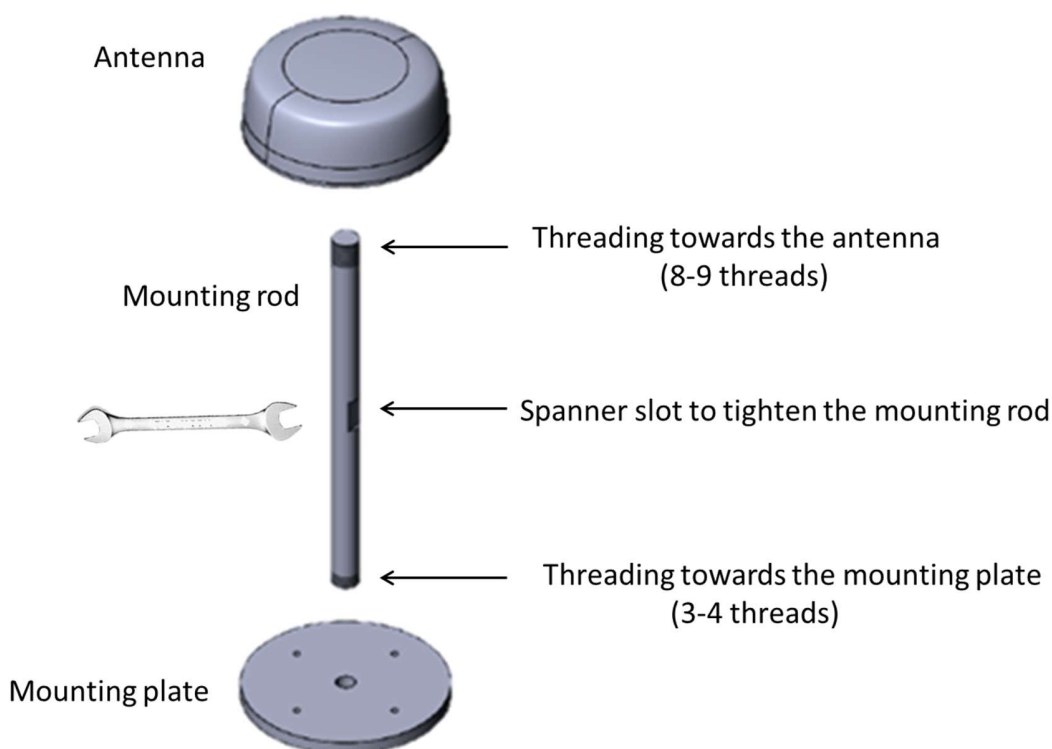


FIGURE 3: ANTENNA MOUNTING PROCEDURE

- Insert one end of the antenna mounting rod (the end which has 3 to 4 threads only) in to the slot provided on the mounting plate and tighten using the spanner supplied until it gets tightly secured
- Fix the mounting plate on any flat surface like concrete/metallic structure on the rooftop or on land vehicle using four slots provided on the mounting plate
- Insert the pole mount Adapter on the antenna in to the other end of the mounting rod and slowly rotate the antenna to tighten it. Care must be taken not to overtighten the thread as it might damage the antenna

3.1.1 ANTENNA MOUNTING CONSIDERATIONS:

- Antenna should be mounted vertically on a flat surface such that there are no obstacles like wall or any other superstructure (as shown in Figure 4) which can block the signal transmitted from satellite to reach the antenna.

- Care should be taken to mount the antenna in vertical direction (approximately perpendicular to the horizon) to avoid antenna gain variations
- The antenna should not be installed close to any source of Electro Magnetic Interference (EMI) or a magnetic compass.
- It is advisable to use lightening protection devices to pass on large portion of the transient energy to ground (earth) during stormy/lightning conditions. This can protect the receiver from getting damaged.

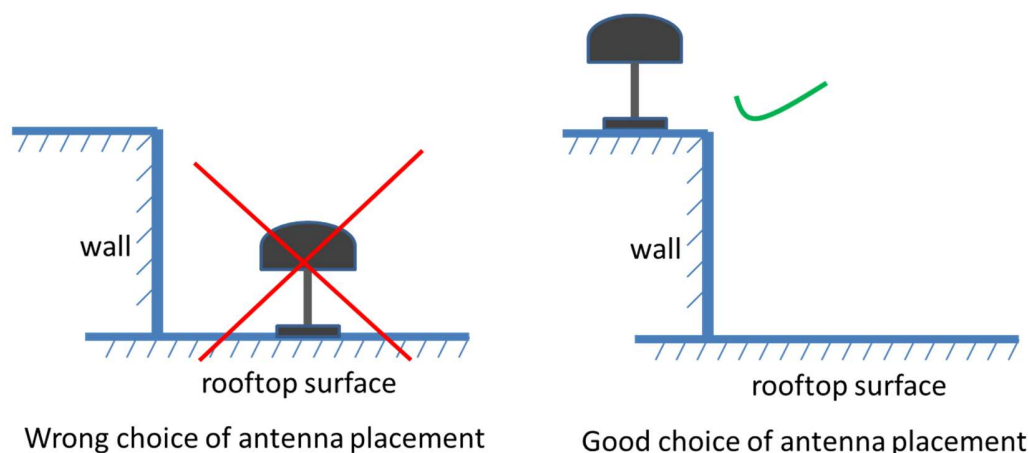


FIGURE 4: ANTENNA MOUNTING CONSIDERATIONS

3.2 POWER INPUT

TWSTFT MODEM can be powered from an AC input, a DC input or with both AC, and DC input (DC input is an option). Supplying both AC and DC input power provides redundant and automatic power switchover in case one or the other input power sources is lost.

AC input acts as the primary power source when both AC and DC are connected.

3.2.1 USING AC INPUT POWER

Connect the AC power cord supplied, to the AC input on the rear panel and the AC power source Socket which supports 230V, 50Hz.

Power ON the AC input by the switch provided in the rear panel of the TWSTFT MODEM.

3.2.2 USING DC INPUT POWER

Connect the DC Cable supplied, to the DC input on the rear panel and the DC power source.

TWSTFT MODEM operates at a range of 24 V- 32V and minimum supply current rating of 2A.

Power ON the DC input by the switch provided on the rear panel of the TWSTFT MODEM.

Before connecting power to the unit, be sure that you have read all safety information.

3.3 MODEM SETUP

- Install the antenna using the antenna setup guidelines as described earlier in section Antenna Setup.
- Connect the antenna to the MODEM using the supplied 15 meter or 2 meter TNC low loss RF cable as shown below.
- Power the MODEM using either: AC input or DC input as described earlier in section Power Input. The power cable from any of the power sources mentioned above can be connected to the MODEM as shown below
- Ethernet cable should be connected between the MODEM and laptop/PC as shown below

3.3.1 POWERING ON

- The MODEM is powered on using the ON/OFF switch of AC or DC input. When the MODEM is switched ON, the LED in the front panel of MODEM should be ON (Power LED will glow).
- Also the LCD should turn ON providing the product information and changing the LCD screen to home screen.

3.3.2 SETTING UP IP ADDRESS

In order for TWSTFT MODEM to be accessible via your network, you need to assign an IP address to MODEM, as well as a subnet mask and gateway, unless you are using an address assigned by a DHCP server.

- **Assigning a Static IP Address**

There are two ways to assign Static IP address

- 1. Through front panel Keypad:**

Foremost step is to disable DHCP in LAN-1 Settings using keypad & LCD.

- Press ENTER key and navigate to LAN ports page through Arrow keys.
- Go to LAN ports -> LAN -1 setting -> DHCP using navigation keys in front panel.
- Disable DHCP and press ENTER key.

Note: TWSTFT MODEM is provided with a default IP address: 192.168.0.27 with subnet mask: 255.255.252.0

- Assign a static IP Address of same subnet or you can retain the default IP address (i.e. 192.168.0.27).

Note: If the entered values are not configured, it displays the previous values.

- Enter subnet mask and Gateway address.
- On successful assignment of new static IP address to TWSTFT MODEM, enter the IP address of MODEM to any WEB browser (Chrome, Firefox) in the URL Tab.

- 2. Through WEB UI via Ethernet LAN-1**

Foremost step is to disable DHCP in LAN-1 Settings using keypad in LCD.

- A. Press ENTER key in front panel of the MODEM and navigate to LAN ports page through Arrow keys.
- B. Go to LAN ports -> LAN-1 setting -> DHCP using navigation keys in front panel.
- C. Disable DHCP and press ENTER or 'o' key.
- D. Default IP address is displayed in IP address section. (i.e. 192.168.0.27) in LCD.
- E. Enter the default IP address (i.e. 192.168.0.27) in URL Tab of WEB browser.
- F. Communication sets up between the workstation and TWSTFT MODEM.
- G. Login and go to LAN settings page in WEB UI.
- H. Click on 'MODIFY' and enter the new IP address in LAN-1 settings and Click on 'SUBMIT' button.
- I. Connection between MODEM and Workstation would be lost.

- J. Enter the newly assigned IP address in URL tab of WEB browser. Communication sets up between TWSTFT MODEM and workstation.
- K. You will be directed to Login page.

- **Using DHCP server**

DHCP server automatically assigns an IP address to MODEM when connected to the network.

To know the assigned IP address of MODEM:

- A. Press ENTER key in front panel of the MODEM and navigate to LAN ports page through Arrow keys.
- B. Go to LAN ports -> LAN-1 setting using navigation keys in front panel.
- C. IP address, subnet mask and gateway assigned by DHCP server is displayed.
- D. Go to PC/Laptop which is connected to the network. Open WEB browser like chrome or Firefox.
- E. Enter the IP address assigned by DHCP server in URL tab.
- F. Communication sets up between PC/Laptop and MODEM.
- G. You will be directed to Login page.

3.4 ACCESSING THE WEB UI

TWSTFT MODEM Web UI is the recommended tool to interact with the device, since it provides access to all configurable settings, and obtain comprehensive status.

You can access the Web UI either by using the automatically assigned DHCP IP address, or by using a manually set static IP address (refer section Setting up IP Address):

1. On a computer connected to the TWSTFT MODEM network, start a web browser, and enter the IP address shown on the LCD front panel of MODEM in LAN-1 Settings page, or setup manually beforehand into the web browser address.
2. Log into the Web UI as an administrator. The factory-default administrator User name & Password is:

Username: accord

Password: accord

Note: It is recommended to change the default Username & Password for security reasons.

4 CONFIGURATION & OPERATION THROUGH LCD

4.1 TWSTFT

4.1.1 STATUS

1. **Mode:** Gives the information of the MODEM for which mode it has been configured.
 - a. Master
 - b. Slave

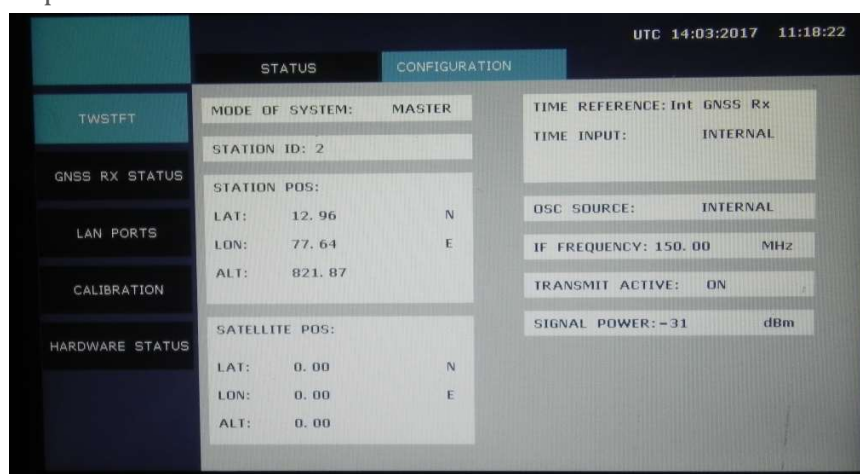


Note: On power ON MODEM will be configured as Slave(i.e. Default Settings)

2. **Time Init:** Gives time initialization information.
 - a. Yes
 - b. No
3. **Sync Status:** Gives information about Synchronization status.
 - a. Not Sync
 - b. Sync
 - c. Holdover
4. **Last Synchronized:** Gives information about last Synchronized time
5. **Channel Status:**
6. **Master Position:**

4.1.2 CONFIGURATION

1. **Mode of System:** It describes operation of the MODEM, whether it is configured as SLAVE or MASTER. To change the mode, navigate to TWSTFT Tab through Navigation keys
 -> press ENTER or 'O' key
 -> select the required mode using UP ARROW or DOWN ARROW key
 -> press ENTER or 'O' to confirm.



2. **Station ID:**
3. **Station Pos:**
4. **Satellite Pos:**
5. **Time Reference:**
6. **Osc Source:**
7. **IF Frequency:** To configure IF Frequency which ranges from 142 MHz to 158 MHz. To change the frequency level, navigate to TWSTFT Tab through navigation keys select IF Frequency parameter
 -> press ENTER or 'O' key
 -> Cursor gets displayed
 -> Enter the value within the specified range
 -> press ENTER or 'O' to confirm.
8. **Transmit Active:** To start or stop transmitting the IF signal.
Note: During Transmit Active ON, no other parameters can be configured except Signal Power parameter.

To configure other parameters in this page, make sure you have turned OFF Transmit Active.

- 9. Signal Power:** To configure the output IF signal power, which ranges from 0 to -40dBm. To change the power level, navigate to TWSTFT Tab through navigation keys select Signal Power parameter
- > press ENTER or 'O' key
 - > Cursor gets displayed
 - > Enter the value within the specified range
 - > press ENTER or 'O' to confirm.

4.2 GNSS RX

4.2.1 CHANNEL DATA

1. UTC Time:
2. TTFF: Time To First Fix
3. TSPE:
4. Clock Bias:
5. Position:
6. PDOP:
7. TDOP:

4.2.2 GPS

1. Channel No:
2. PRN ID:
3. C/NO:
4. STATUS:
5. IONO TYP:

4.2.3 IRNSS L5

1. Channel No:
2. PRN ID:
3. C/NO:
4. STATUS:

5. IONO TYP:

4.2.4 IRNSS

1. **Channel No:**
2. **PRN ID:**
3. **C/NO:**
4. **STATUS:**
5. **IONO TYP:**

4.2.5 GLONASS

1. **Channel No:**
2. **PRN ID:**
3. **C/NO:**
4. **STATUS:**
5. **IONO TYP:**

4.2.6 SBAS

1. **Channel No:**
2. **PRN ID:**
3. **C/NO:**
4. **STATUS:**
5. **IONO TYP:**

4.2.7 CONFIGURATION

1. **Mode of Positioning:**
 - Pos Fix Mode:**
 - Survey Mode:**
2. **Position Fix:**
3. **LNA Delay:** This command is used to program/compensate the LNA delay.
4. **RESET:**

5. **Cable Delay:** This command is used to program/compensate the antenna cable delay.
6. **RF Delay:** This command is used to program/compensate the RF delay.

4.3 LAN PORTS

4.3.1 LAN-1 SETTINGS

1. **IP Address:**
2. **Subnet Mask:**
3. **Gateway:**
4. **DHCP:**
5. **SNMP Trap IP:** Enter SNMP Manager IP address for which the agent (TWSTFT MODEM) is going to be monitored for status messages.
6. **Link Status:**

4.3.2 LAN-2 SETTINGS

1. **IP Address:**
2. **Subnet Mask:**
3. **Gateway:**
4. **DHCP:**
5. **Link Status:**

4.3.3 PTP

1. **PTP Enable:** This command is used to Enable/ Disable PTP.
2. **STEP Mode:** This is used to select mode of operation.
 - a. **One Step Mode**

PTP supports two different delay mechanisms each with step modes. Step mode indicates support for adding timestamp within the packet or in the following packet.

In End to End Mode, One step indicates Sync message will contain timestamp and there will be no Follow_Up message.

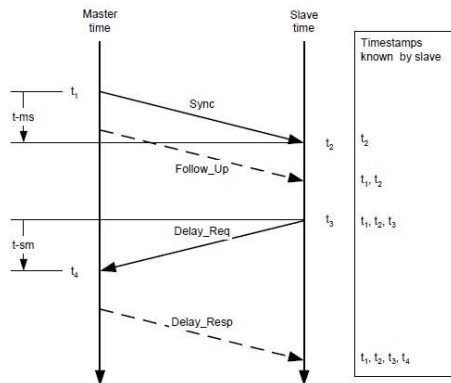


FIGURE 5: END TO END DELAY MECHANISM

Here dotted lines indicate the packets are not time critical (General Messages).
Thick lines indicate packets are time critical (Event Messages).

In Peer to Peer Mode, One step indicates Pdel_Resp contains timestamp and there will be no Pdel_Resp_Follow_Up message

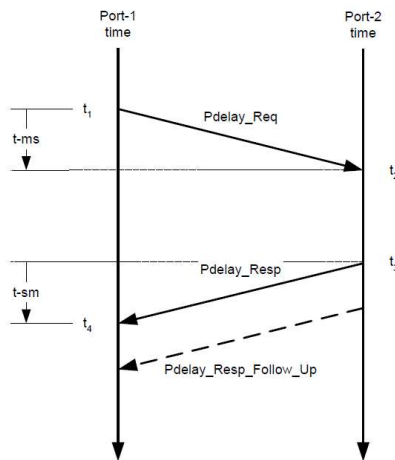


FIGURE 6: PEER TO PEER DELAY MECHANISM

b. Two Step Mode

In End to End Mode, Two Step indicates Sync message timestamp will be there in Follow_Up message.

In Peer to Peer Mode, Two Step indicates Pdel_Resp timestamp will be there in Pdel_Resp_Follow_Up message.

3. **Time to Live:** This field is used to select the TTL of the packet indicating number of switches the packets can cross.
4. **Domain No:** This field defines the subdomain number of the PTP network.
5. **Priority-1:** This field is used for the BMC (Best Master Clock) algorithm.
6. **Priority-2:** This field is used for the BMC (Best Master Clock) algorithm.
7. **Delay Mechanism:** This field is used to select the delay calculation mechanism.
 - a. **End to End**

In End to End Delay Mechanism, delay measurements are done between to end devices, assuming the delay in both directions in network to be same. This is majorly used in networks where PTP compliant switches are missing.

- b. **Peer to Peer**

In Peer to Peer Delay Mechanism, delay measurements are done between peer devices, assuming the delay measured in cables be same in both the directions. Compared to End to End DM this assumption holds better unless the cable is very long provided PTP compliant switches are used over the network.

8. **Restart:** This is to restart the PTP operations.
9. **Status:**
 - a. **PORT STATE:** This indicates the PTP port is in what state.
 - i) **Initializing**
 - ii) **Master**
 - b. **PTP Version:** This displays the current version of PTP being supported as 2 (IEEE 1588-2008, v2)
 - c. **IDENTITY:** This indicates 8 bytes of clock identity of PTP device. This will be a unique Identification number for each PTP Clock used to differentiate message sources and destination.
 - d. **UTC OFFSET:** The current UTC Offset w.r.t., TAI (Temps Atomic International, is the international atomic time).
 - e. **PORT NUMBER:** This is a hardware port number of a PTP device.
 - f. **PTP TIME SOURCE:** This indicates the time source of PTP device
 - i. **GPS**
 - ii. **Internal Oscillator**
 - iii. **NTP**

4.3.4 NTP

The Network Time Protocol (NTP) is widely used to synchronize computer clocks in the network. NTP is UDP based server-client protocol.

NTP server will synchronizes to a specific clock source and serves other NTP peers.

TWSTFT Modem acts as NTP Primary server and serves its peers requests via LAN-2 port. Modem supports NTP v3 and v4 with symmetric key security feature.

TWSTFT-NTP primary server configurations can be altered from LAN-2 NTP page. Following section will describe various features along with the descriptions.

1. **NTP Enable:** This command is used to Enable/ Disable NTP.
Note: NTP is enabled on Power ON.
2. **Broadcast:** This command is used to Enable/Disable Broadcast message.
Modem Broadcasts NTP message to the network with 255.255.255.0 ip address as a primary stratum NTP server. This NTP broadcast message showcase the NTP primary server availability in the network.
Note: NTP Broadcast will be enabled on Power ON.
3. **Multicast:** This command is used to Enable/Disable Multicast message.
Modem multicasts NTP message to the network with I.P address 224.0.1.1 as a primary stratum NTP server. This NTP multicast message showcase the NTP primary server availability in the network.
4. **Broadcast Interval:** This command defines the interval of Broadcast message.
Broadcast interval defines the periodicity of broadcast message transmission over the network. Interval will be in log2 seconds format.
Note: Broadcast interval is 6(64 Seconds) on Power ON.
5. **Multicast Interval:** This command defines the interval of Multicast message.
Multicast interval defines the periodicity of multicast message transmission over the network. Interval will be in log2 seconds format.
6. **Status:**
This tab shows the status of NTP server state variable like NTP server mode, Version number, Stratum, Leap indicator, Clock source, Frequency offset and Jitter.

TWSTFT Modem configured as MASTER and time is initialized.	
Time Reference	NTP Mode
Internal GNSS Receiver	SERVER
Manual	SERVER
NTP Server	CLIENT. Once the time initialization is done in the TWSTFT Modem then NTP mode becomes SERVER
External GNSS Receiver	NA
Master	NA

TWSTFT Modem configured as SLAVE and time is initialized.	
Time Reference	NTP Mode
Internal GNSS Receiver	NA
Manual	NA
NTP Server	NA
External GNSS Receiver	SERVER
Master	SERVER

Note: In MASTER/SLAVE configurations if time is not initialized, then NTP mode will be represented as “INIT” (Initialization).

NTP Version: Represents modem NTP version number, currently 4.

Stratum: Defines the class of the source clock.

Example: “Primary server” (e.g., equipped with a GPS receiver)

Leap Indicator: It is a warning of an impending leap-second to be inserted or deleted in the last minute of the current month.

Reference Clock Source: Server ID or Clock Reference Identifier which is maintained by IANA.

Jitter: Indicator of fundamental timekeeping performance and network congestion state.

Frequency Offset: Represents the frequency offset of clock source.

***Time offset:** Maximum-likelihood time offset of the server clock relative to the system clock.

***Root delay:** Round-trip delay between the client and server.

Note: The parameters with start (*) suffix, will be displayed on the LCD/webpage in the NTP Server time initialization mode.

5 CALIBRATION

1. Cable Delay:

- a. **RS-422 1 PPS:** This command is used to program/compensate RS-422 1 PPS cable delay.
- b. **RS-422 10 MHz:** This command is used to program/compensate RS-422 10 MHz cable delay.
- c. **TTL 1 PPS:** This command is used to program/ compensate TTL 1 PPS cable delay.
- d. **GNSS Antenna:**

2. Offset w.r.t Master:

- a. **TWSTFT Slave:**

6 HARDWARE STATUS

- 1. **Optical Signal Out:**
- 2. **SD Card:** To Mount & Unmount SD card
- 3. **SD Card Log:** To Start & Stop logging data in SD card
- 4. **Time Since Power ON:** It gives the count in seconds, since the MODEM was turned ON.
- 5. **INT OSC Type:** It gives info of the type of Internal Oscillator used in the MODEM.
- 6. **INT OSC CNTRL Voltage:**
- 7. **Ext CLK PWR:**
- 8. **Optical Power Type:**

7 CONFIGURATION & OPERATION THROUGH WEB UI

This section explains the usage of WEB UI application and how to configure the TWSTFT/NGS T20 device. It allows the user to view device status, configure mode of system, and restart services like NTP, PTP, and Hardware etc.

5.1 OPERATING WEB UI

Launch the WEB UI by providing the IP address of device in URL (Ex: 192.168.1.2). After that, Sign In page will appear as shown in Figure 7-1 where user can enter the username and password. (Note: Compatible browser: Mozilla Firefox and Google Chrome – version 40 and above).

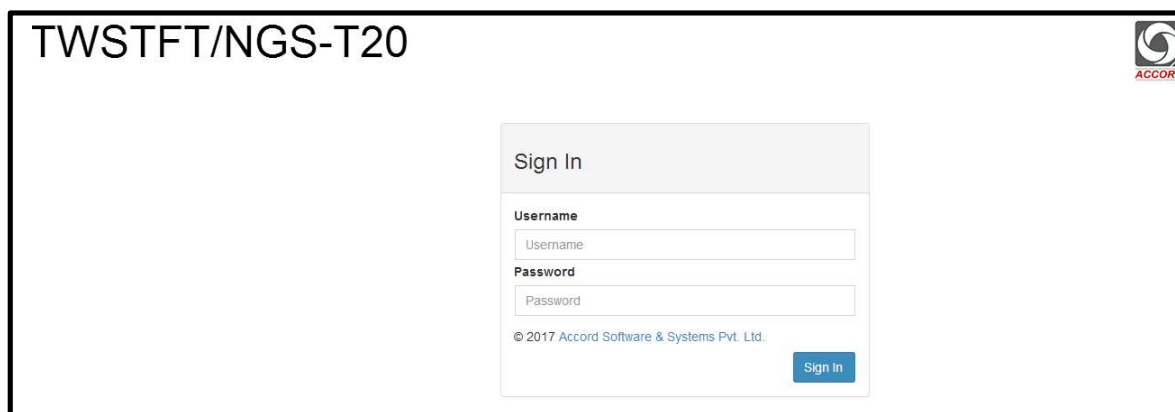


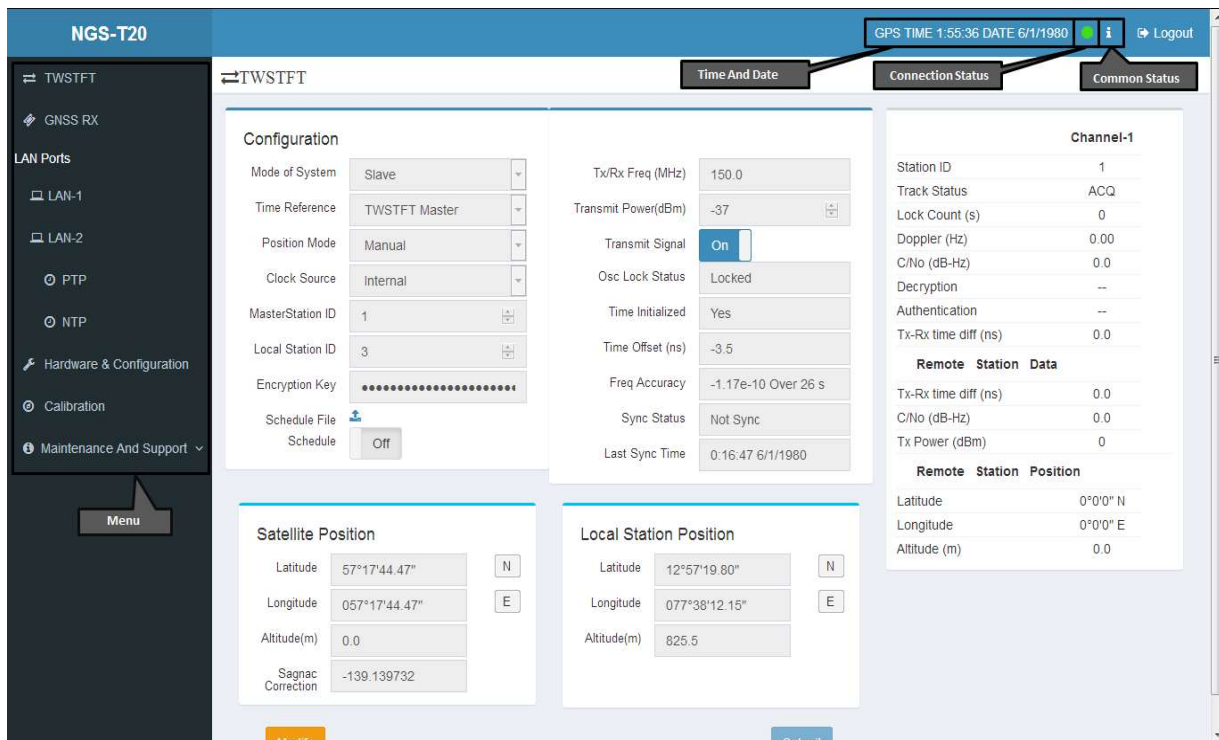
FIGURE 7-1 SIGN IN

The common features that are present in all web pages are highlighted in Figure 7-2 and described as follows

- MENU – Displays the features available in application.
- Time and Date – Displays the device time, date and time format (UTC or GPS) information.
- Logout
- Connection Status-Indicates hardware connection status

Green – Connected Red – Disconnected

- Common Status –Indicates status of hardware services (like NTP and PTP) and LAN ports as shown in Figure 7-3.



NGS-T20

GPS TIME 1:55:36 DATE 6/1/1980 Logout

TWSTFT

Configuration

Mode of System: Slave

Time Reference: TWSTFT Master

Position Mode: Manual

Clock Source: Internal

Master Station ID: 1

Local Station ID: 3

Encryption Key:

Schedule File: +

Schedule: Off

Tx/Rx Freq (MHz): 150.0

Transmit Power(dBm): -37

Transmit Signal: On

Osc Lock Status: Locked

Time Initialized: Yes

Time Offset (ns): -3.5

Freq Accuracy: -1.17e-10 Over 26 s

Sync Status: Not Sync

Last Sync Time: 0:16:47 6/1/1980

Channel-1

Station ID: 1

Track Status: ACQ

Lock Count (s): 0

Doppler (Hz): 0.00

C/No (dB-Hz): 0.0

Decryption: --

Authentication: --

Tx-Rx time diff (ns): 0.0

Remote Station Data

Tx-Rx time diff (ns): 0.0

C/No (dB-Hz): 0.0

Tx Power (dBm): 0

Remote Station Position

Latitude: 0°0'0" N

Longitude: 0°0'0" E

Altitude (m): 0.0

Satellite Position

Latitude: 57°17'44.47" N

Longitude: 057°17'44.47" E

Altitude(m): 0.0

Sagnac Correction: -139.139732

Local Station Position

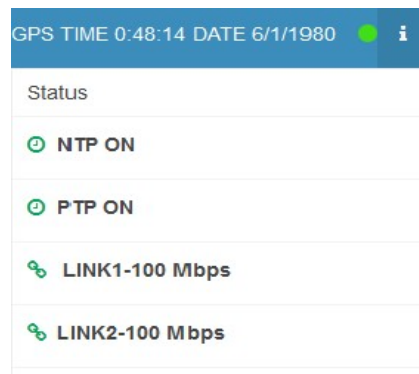
Latitude: 12°57'19.80" N

Longitude: 077°38'12.15" E

Altitude(m): 825.5

Modify Submit

FIGURE 7-2: COMMON LAYOUT



GPS TIME 0:48:14 DATE 6/1/1980 Logout

Status

ON **NTP ON**

ON **PTP ON**

100 Mbps **LINK1-100 Mbps**

100 Mbps **LINK2-100 Mbps**

FIGURE 7-3 : COMMON STATUS

5.2 TWSTFT

Clicking on TWSTFT option in menu, a page will appear as shown in Figure 7-4. “Modify” button allows user to configure the fields present in TWSTFT page. “Submit” button will transmit the modified data to device.

NGS-T20

GPS TIME 2:40:30 DATE 6/1/1980 Logout

TWSTFT

GNSS RX

LAN Ports

LAN-1

LAN-2

PTP

NTP

Hardware & Configuration

Calibration

Maintenance And Support

TWSTFT

Configuration

Mode of System

Slave

Time Reference

TWSTFT Master

Position Mode

Manual

Clock Source

Internal

MasterStation ID

1

Local Station ID

3

Encryption Key

.....

Schedule File

+

File

Off

Tx/Rx Freq (MHz)

150.0

Transmit Power(dBm)

-37

Transmit Signal

On

Osc Lock Status

Locked

Time Initialized

Yes

Time Offset (ns)

-3.5

Freq Accuracy

-1.17e-10 Over 26 s

Syno Status

Not Sync

Last Sync Time

0:16:47 6/1/1980

Channel-1

Station ID

1

Track Status

ACQ

Lock Count (s)

0

Doppler (Hz)

0.00

C/No (dB-Hz)

0.0

Decryption

--

Authentication

--

Tx-Rx time diff (ns)

0.0

Remote Station Data

Tx-Rx time diff (ns)

0.0

C/No (dB-Hz)

0.0

Tx Power (dBm)

0

Remote Station Position

Latitude

0°0'0" N

Longitude

0°0'0" E

Altitude (m)

0.0

Satellite Position

Latitude

57°17'44.47"

N

Longitude

057°17'44.47"

E

Altitude(m)

0.0

Sagnac Correction

-139.139732

Local Station Position

Latitude

12°57'19.80"

N

Longitude

077°38'12.15"

E

Altitude(m)

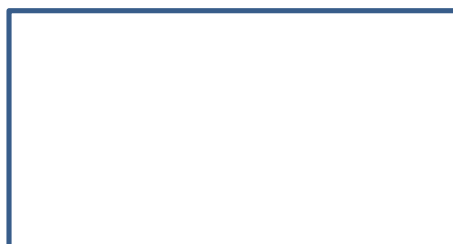
825.5

Modify

Submit

FIGURE 7-4: TWSTFT PAGE

Once Modified data is transmitted to device, a popup appears indicating “Configuration! - Done” as shown in Figure 7-5.



TWSTFT- USER MANUAL

Page
30

Rev 1.0

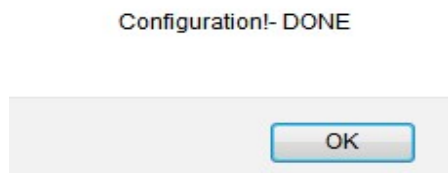


FIGURE 7-5 : CONFIG. DONE

Note:

USER CAN ONLY MODIFY TRANSMIT SIGNAL POWER AND TRANSMIT SIGNAL, IF TRANSMIT SIGNAL IS “ON” AS SHOWN IN

✓ Figure 7-6.

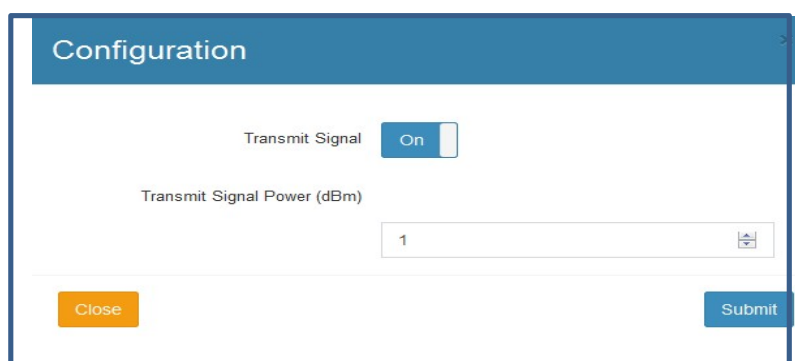


FIGURE 7-6: TRANSMIT STATUS

5.3.1 SCHEDULE FILE

It provides feature for synchronising master with 32 slaves based on duration provided for each slave. At a time master can synchronise to two slaves.

“Download” button beside Download Schedule file heading enables user to download default schedule file present in device.

Steps to upload Schedule File:

- To upload a schedule XML file click on  button in TWSTFT page.
- “Browse” button allows user to select schedule file.
- Once schedule file is selected, its details will be displayed, as shown in Figure 7-8.

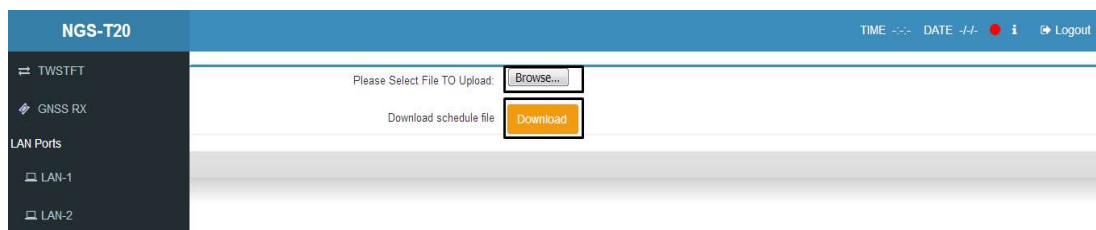
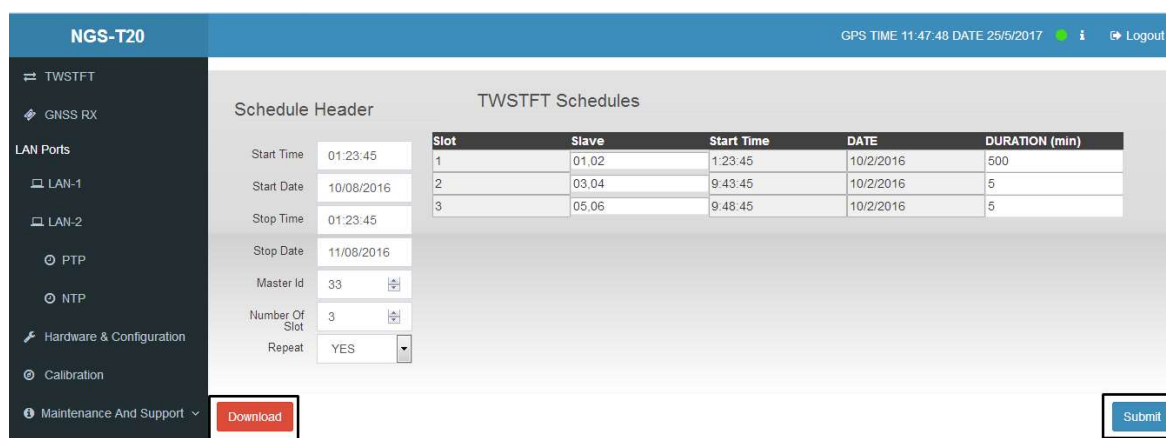


FIGURE 7-7: SCHEDULE BROWSE



Slot	Slave	Start Time	DATE	DURATION (min)
1	01,02	1:23:45	10/2/2016	500
2	03,04	9:43:45	10/2/2016	5
3	05,06	9:48:45	10/2/2016	5

FIGURE 7-8: SCHEDULE DETAILS

Steps To Submit/Download Modified Schedule File:

- “Download” button allows user to download schedule file in xml format.
- “Submit” button allows user to transmit modified schedule data to device.

Special Case:

1. If mode of system is Master and Time Reference is manual then a dialog pops up for updating date and time as shown in Figure 7-9.



FIGURE 7-9: DATE & TIME POPUP

5.3 GNSS RECEIVER

It contains two tabs i.e. GNSS and Calibration.

GNSS Tab

- In GNSS tab user can modify Receiver position data (i.e. latitude, longitude and altitude) only in Posfix mode under mode of positioning field.
- “Submit” button transmits modified position data to device.
- “Receiver Reset” button resets receiver.
- It also displays channel status of constellations like GPS, GLONASS, IRNSS and SBAS.

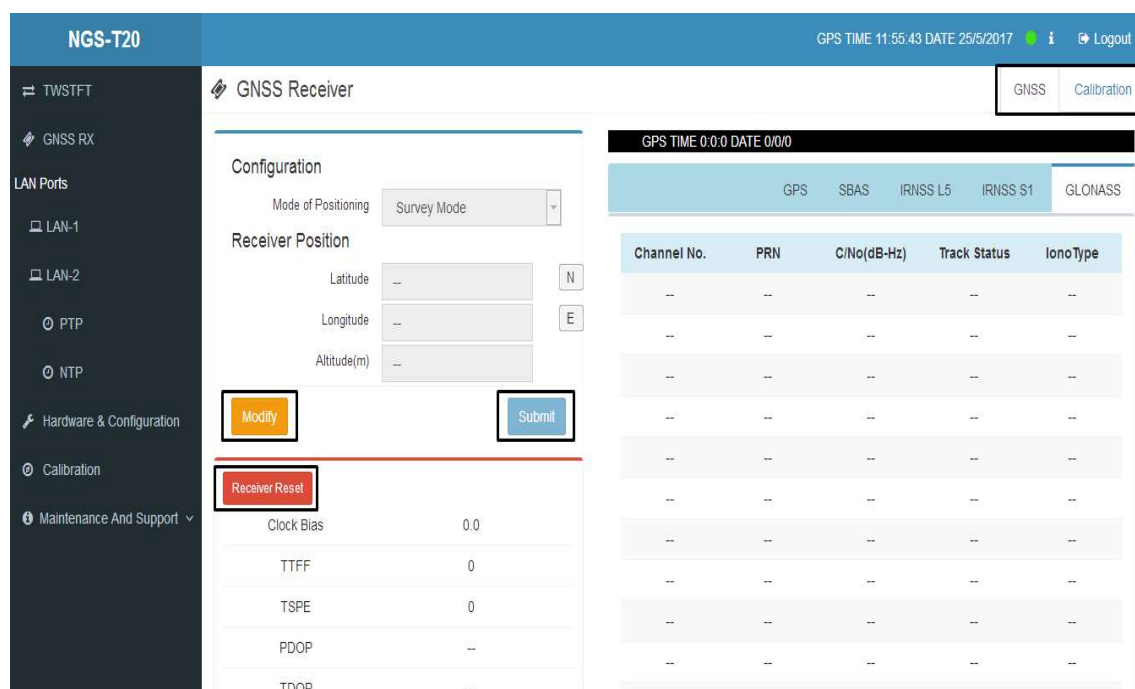


FIGURE 7-10: GNSS RECEIVER

Calibration Tab:

- This page provides detail of cable, LNA and RF delay of GPS, GLONASS and IRNSS.
- “Modify” button popups a dialog to configure delay.
- Select constellation, delay type and provide delay.
- “Submit” button transmits modified data to device.

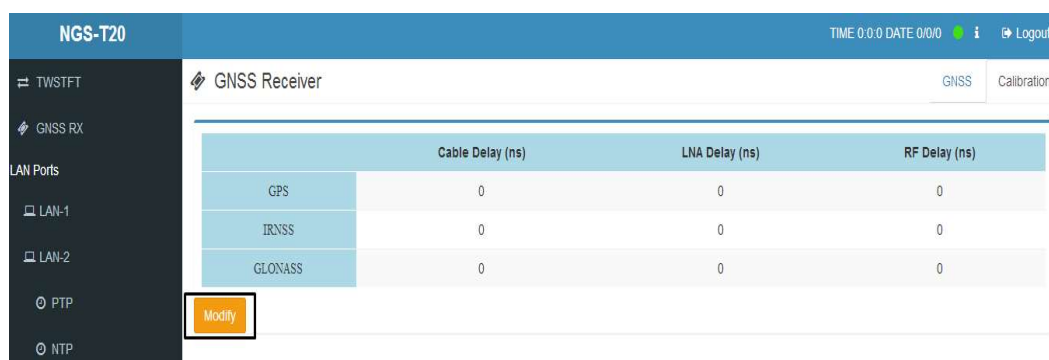


FIGURE 7-11: GNSS -CALIBRATION

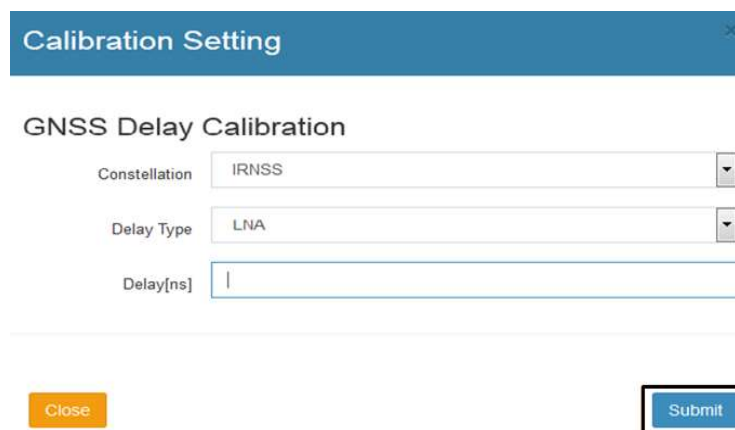


FIGURE 7-12: GNSS –CALIBRATION SETTING

5.4 LAN1 AND LAN2

- It provides LAN port configurations like IP address, subnet mask, gateway, DHCP status, Link status and Trap IP.
- “Modify” button enables user to configure port settings.
- Invalid fields are highlighted with red colour as shown in Figure 7-14.
- “Submit” button transmits valid port configuration to device as shown in Figure 7-13.
- “LAN2-Reset” button resets LAN2 port.

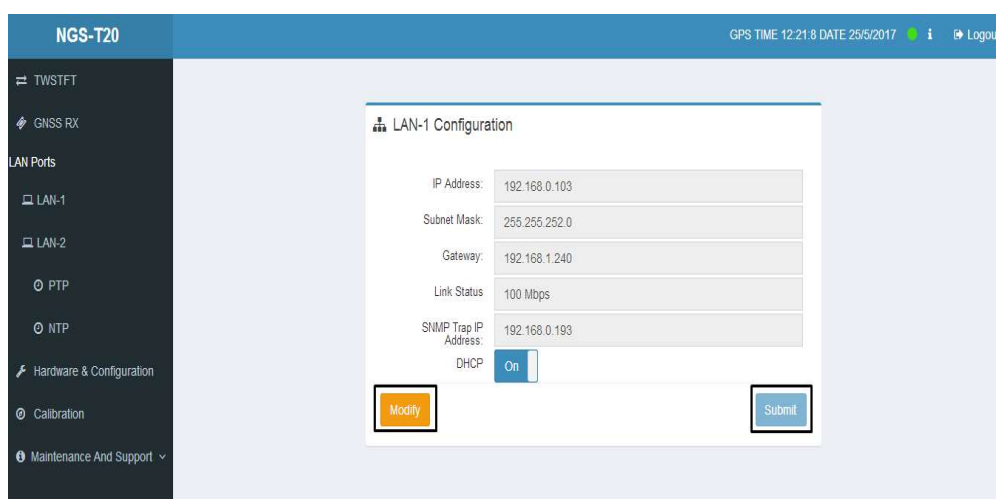
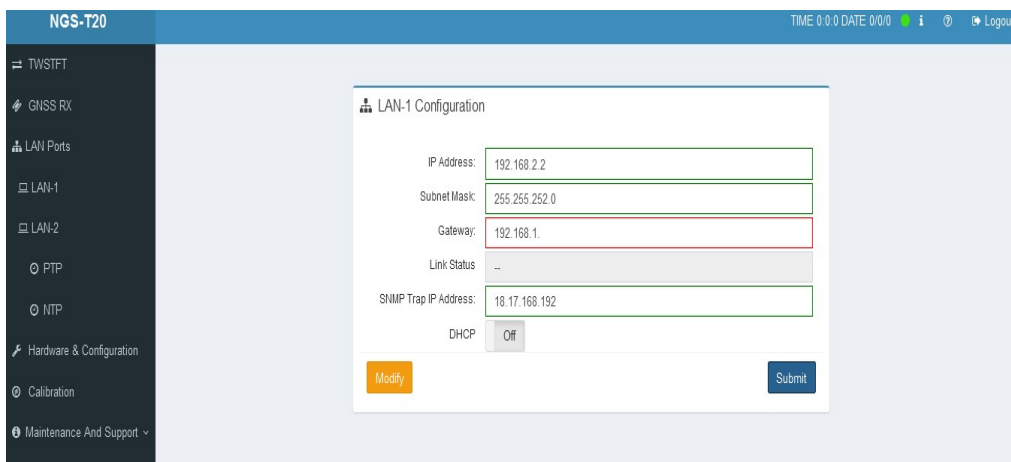


FIGURE 7-13: LAN CONFIGURATION

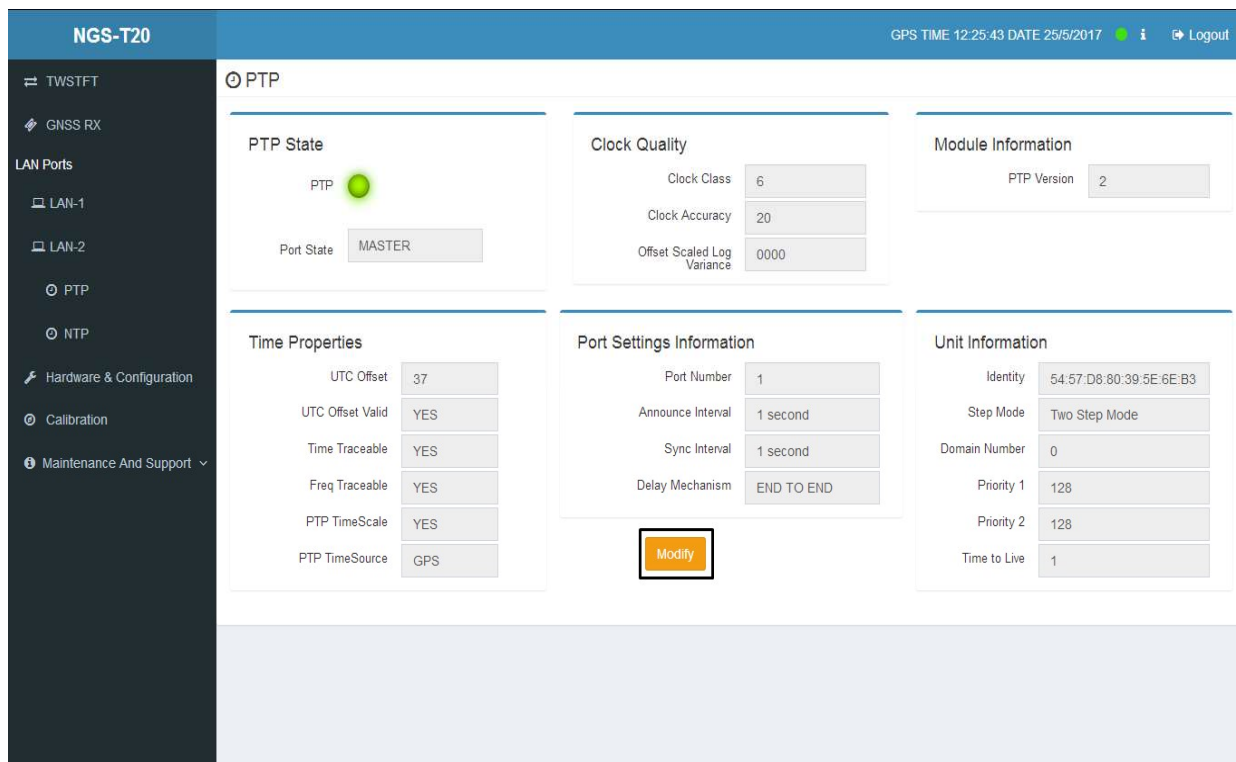


The screenshot shows the NGS-T20 web interface. The left sidebar contains a menu with options: TWSTFT, GNSS RX, LAN Ports, LAN-1, LAN-2, PTP, NTP, Hardware & Configuration, Calibration, and Maintenance And Support. The main content area displays the 'LAN-1 Configuration' form. The form includes fields for IP Address (192.168.2.2), Subnet Mask (255.255.252.0), Gateway (192.168.1), Link Status (--), and SNMP Trap IP Address (18.17.168.192). There is a DHCP toggle set to 'Off'. At the bottom of the form are 'Modify' and 'Submit' buttons. The top right of the interface shows 'TIME 0:0:0 DATE 0/0/0' and a 'Logout' link.

FIGURE 7-14: LAN VALIDITY

5.5 PTP

- PTP page displays PTP configuration of device (Figure 7-15).



The screenshot shows the NGS-T20 web interface with the 'PTP' option selected in the left sidebar. The main content area displays the 'PTP Status' page. The page is divided into several sections:

- PTP State:** Shows a green 'PTP' indicator and 'Port State' as 'MASTER'.
- Clock Quality:** Shows 'Clock Class' as 6, 'Clock Accuracy' as 20, and 'Offset Scaled Log Variance' as 0000.
- Module Information:** Shows 'PTP Version' as 2.
- Time Properties:** A table with the following values:

UTC Offset	37
UTC Offset Valid	YES
Time Traceable	YES
Freq Traceable	YES
PTP TimeScale	YES
PTP TimeSource	GPS
- Port Settings Information:** A table with the following values:

Port Number	1
Announce Interval	1 second
Sync Interval	1 second
Delay Mechanism	END TO END
- Unit Information:** A table with the following values:

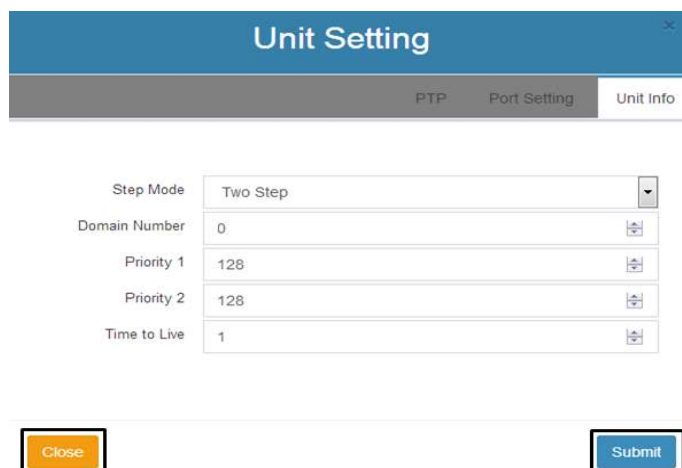
Identity	54:57:D8:80:39:5E:6E:B3
Step Mode	Two Step Mode
Domain Number	0
Priority 1	128
Priority 2	128
Time to Live	1

 A 'Modify' button is located at the bottom of the 'Port Settings Information' section. The top right of the interface shows 'GPS TIME 12:25:43 DATE 25/5/2017' and a 'Logout' link.

FIGURE 7-15: PTP STATUS

- “Modify” button pops up Port setting dialog, it enables user to configure fields in any three tabs i.e. PTP, Port Setting and Unit Info.

- “Submit” button validates and transmits modified data to device as shown in Figure 7-16. If any field is invalid then it will be highlighted with red colour.

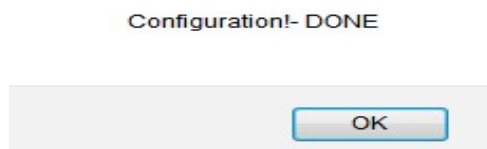


The image shows a 'Unit Setting' dialog box with a blue header and a close button. Below the header are three tabs: 'PTP', 'Port Setting', and 'Unit Info'. The 'PTP' tab is selected. The settings are as follows:

Field	Value
Step Mode	Two Step
Domain Number	0
Priority 1	128
Priority 2	128
Time to Live	1

At the bottom of the dialog are two buttons: 'Close' (orange) and 'Submit' (blue).

FIGURE 7-16: PTP SETTINGS

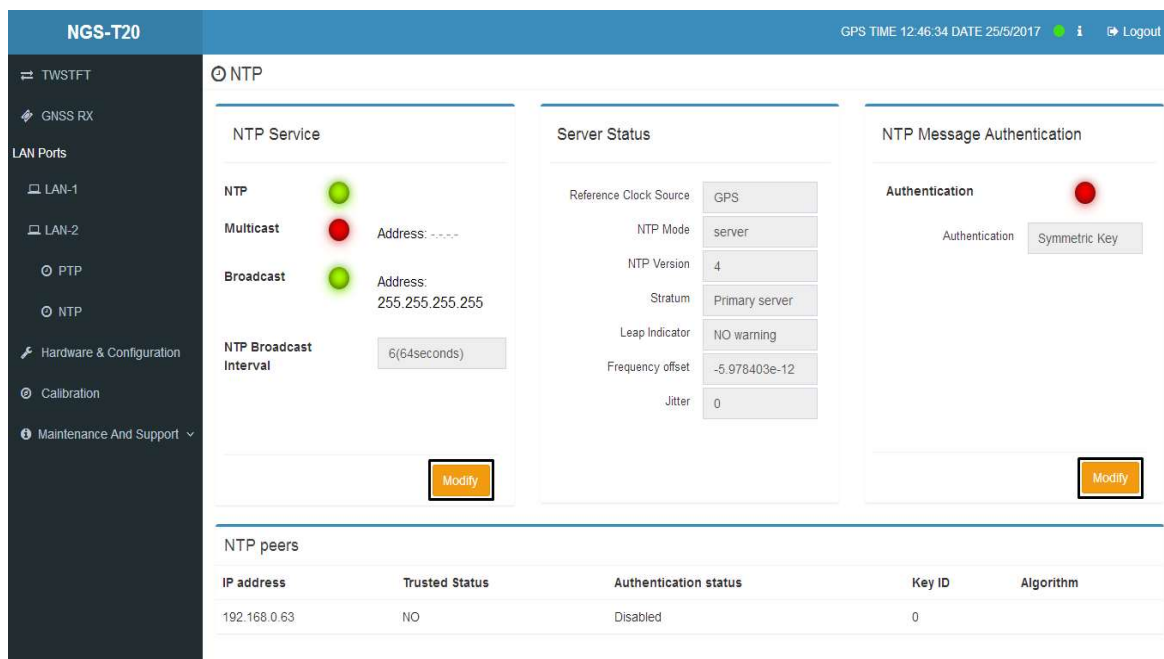


The image shows a 'Configuration!- DONE' dialog box with a light gray background. It contains a single 'OK' button in the center.

FIGURE 7-17: CONFIGURATION

5.6 NTP

- NTP page displays NTP configuration of device, NTP peers (i.e. number of client connected), NTP message authentication and NTP server status (Figure 7-18).



NGS-T20 GPS TIME 12:46:34 DATE 25/5/2017 Logout

NTF

NTP Service

NTP ●

Multicast ● Address: -----

Broadcast ● Address: 255.255.255.255

NTP Broadcast Interval: 6(64seconds)

Modify

Server Status

Reference Clock Source	GPS
NTP Mode	server
NTP Version	4
Stratum	Primary server
Leap Indicator	NO warning
Frequency offset	-5.978403e-12
Jitter	0

NTP Message Authentication

Authentication ●

Authentication: Symmetric Key

Modify

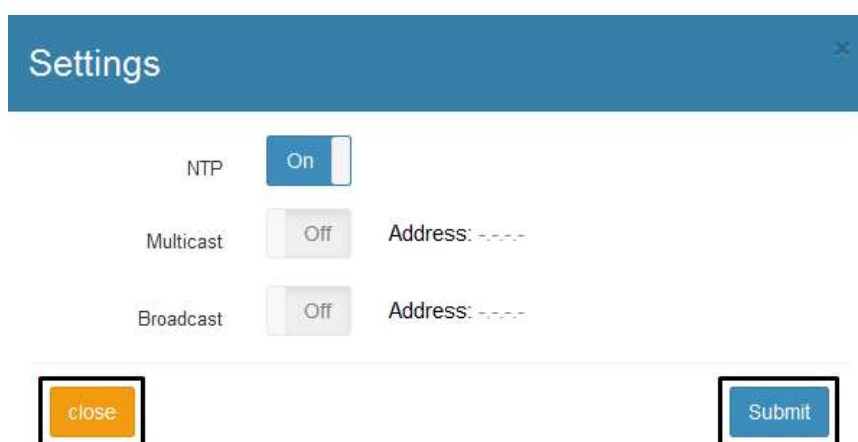
NTP peers

IP address	Trusted Status	Authentication status	Key ID	Algorithm
192.168.0.63	NO	Disabled	0	

FIGURE 7-18: NTP

5.7.1 TO MODIFY NTP SERVICE

User can modify NTP service by clicking on modify button provide under NTP service block. Once modify is clicked a dialog pops up, modified data can be submitted to hardware by clicking on submit button as shown in Figure 7-19.



Settings [X]

NTP On

Multicast Off Address: -----

Broadcast Off Address: -----

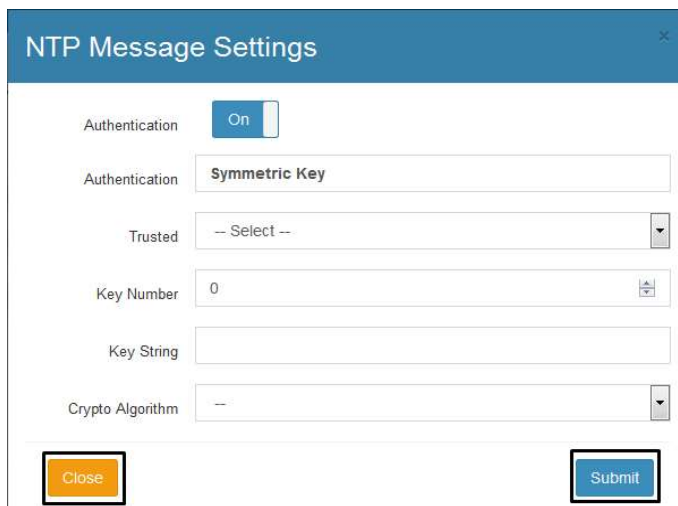
close Submit

FIGURE 7-19: NTP SETTING

5.7.2 TO MODIFY NTP MESSAGE AUTHENTICATION

User can modify NTP Message Authentication by clicking on modify button provided under NTP Message Authentication. Once modify is clicked a dialog pops up, modified data is validated, if it's valid it can be submitted to hardware by clicking on submit button as shown in

Figure 7-20.



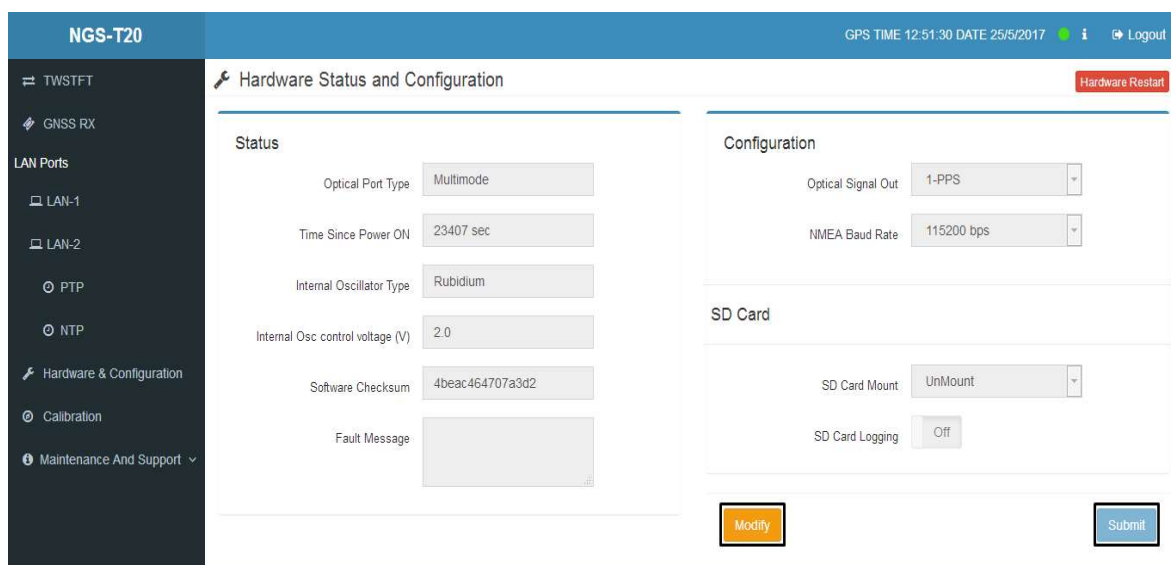
The dialog box titled "NTP Message Settings" contains the following fields and controls:

- Authentication:** A toggle switch set to "On".
- Authentication:** A text input field containing "Symmetric Key".
- Trusted:** A dropdown menu showing "-- Select --".
- Key Number:** A text input field containing "0".
- Key String:** A text input field.
- Crypto Algorithm:** A dropdown menu showing "--".
- Buttons:** "Close" (orange) and "Submit" (blue) buttons at the bottom.

FIGURE 7-20: NTP MESSAGE SETTING

5.7 HARDWARE STATUS AND CONFIGURATION

- This page displays Hardware Status like SD card status, NMEA baud rate etc.
- "Modify" button enables user to configure hardware status.
- "Submit" button will transmit the modified data to device.
- "Restart" button will restart the hardware.



The "Hardware Status and Configuration" page for the NGS-T20 device shows the following sections:

- Header:** NGS-T20, GPS TIME 12:51:30 DATE 25/5/2017, and a Logout link.
- Left Sidebar:** Navigation menu with options: TWSTFT, GNSS RX, LAN Ports (LAN-1, LAN-2), PTP, NTP, Hardware & Configuration (selected), Calibration, and Maintenance And Support.
- Hardware Status and Configuration Section:**
 - Status:**
 - Optical Port Type: Multimode
 - Time Since Power ON: 23407 sec
 - Internal Oscillator Type: Rubidium
 - Internal Osc control voltage (V): 2.0
 - Software Checksum: 4beac464707a3d2
 - Fault Message: (empty text area)
 - Configuration:**
 - Optical Signal Out: 1-PPS
 - NMEA Baud Rate: 115200 bps
 - SD Card:**
 - SD Card Mount: UnMount
 - SD Card Logging: Off
- Buttons:** "Hardware Restart" (red), "Modify" (orange), and "Submit" (blue) buttons.

FIGURE 7-21: HARDWARE STATUS AND CONFIGURATION

5.8 CALIBRATION

- This page displays cable delays.
- “Modify” button enables user to configure cable delays.
- “Submit” button will transmit the modified data to device.

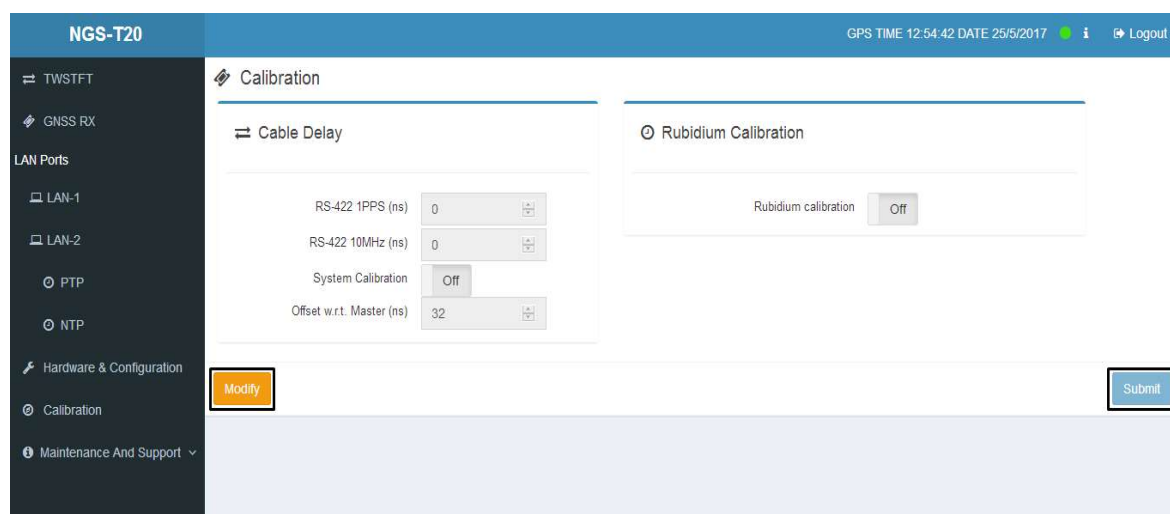


FIGURE 7-22: CALIBRATION

5.9 MAINTENANCE AND SUPPORT

- “Software Upgrade” refer topic (**Error! Reference source not found.**).
- “Contact us” page provides contact details.



Web UI is compatible with Firefox and Google Chrome version above 40.

8 DATA LOGGING THROUGH SD CARD

TWSTFT MODEM data is logged in SD card.

To log data in SD card, make sure SD card is inserted and initialized. If the SD card is inserted before powering ON the unit; on Power on it gets initialized automatically.

If the card is inserted after powering ON the unit; initialization should be done manually. Follow the steps to initialize:

- Navigate to Hardware Status Tab in LCD/WEB Page.
- Select SD card command.
- Choose MOUNT option and press ENTER

To Start /Stop logging data in SD card:

- Navigate to Hardware Status Tab in LCD.
- Select SD card Log command.
- Choose Start/Stop option and press ENTER

	Note: Do not remove SD card while logging. Data logged in the files would be lost. Make sure you have given UNMOUNT command before removing SD Card from the unit.
---	--

Organization of folders in SD card:

Two Separate folders are created, one for TWSTFT parameters and another for GNSS Receiver parameters.

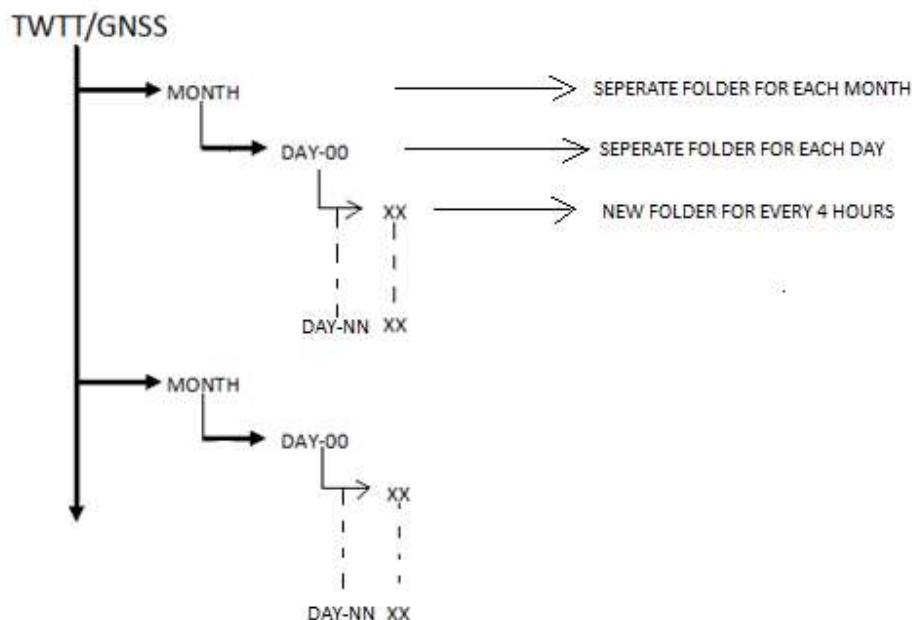


FIGURE 23: FOLDER CREATION LOGIC

8.1 FOLDER CREATION LOGIC:

- TWTT/GNSS folder is created and incremented from the previous logs in SD card for every start & stop of Log.
- Under TWTT/GNSS folder separate folders are created based on the logging date. I.e. separate folder for different month and different day.
- Under the day folder, separate folders are created for every 4 hours
- Inside hour folder TWTT/GNSS file parameters are logged in excel format.

1. TWTT

This folder contains two files namely TWTT.csv & CH-IND.csv

2. GNSS

This folder contains six files namely CHIND.csv, GPS.csv, GLONS.csv, IRNL5.csv, IRNS1.csv & SBAS.csv



Note: Maximum number of TWTT & GNSS folders can be created are 100.(i.e. 0-99)

9 SOFTWARE UPGRADE (SWUG)

Software Upgrade mainly involves two steps:

1. Downloading of latest firmware files from Accord's FTP site.
2. Initiate & perform software upgrade through LCD/WEB page.

User can download the latest firmware files from Accord's FTP site, to the local PC/Laptop.

- Copy the firmware files to SD card provided.
- Insert SD card to the unit and power ON.
- Go to Hardware Status tab in LCD/WEB page.
- Select SWUG command and ENTER.

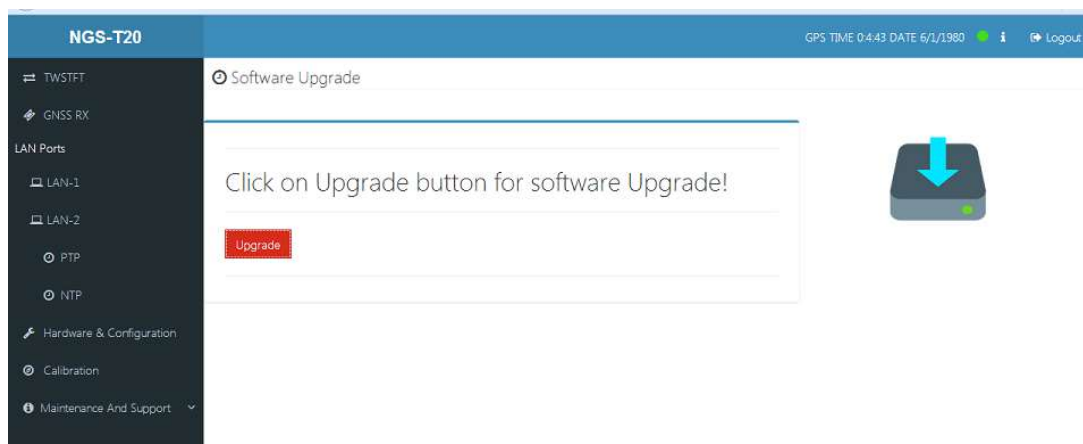


FIGURE 24:SWUG THROUGH WEB UI

Once the command is executed, it enters in to maintenance mode, and starts upgrading with the new firmware files. The status of software upgrade can be seen in LCD.

SPECIFICATIONS



Note: Make sure there is no interruption in input power.

If there is an error while upgrading and has failed to update, TWSTFT MODEM will be in maintenance mode only.

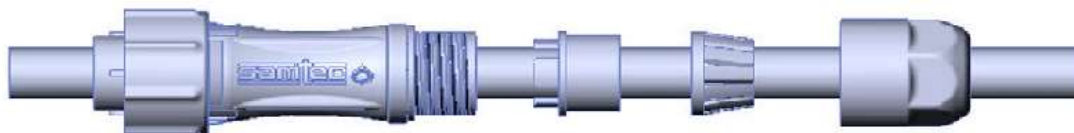
Give a power RESET and start the above process once again.

On successful completion of upgrade, it gives an auto reset to the MODEM and switches from maintenance mode to the application mode and is ready to use now.

10 CABLE DETAILS

10.1 DC-DC INPUT POWER CABLE

TWSTFT MODEM can be powered ON from DC power Source using the supplied DC-DC power cable.

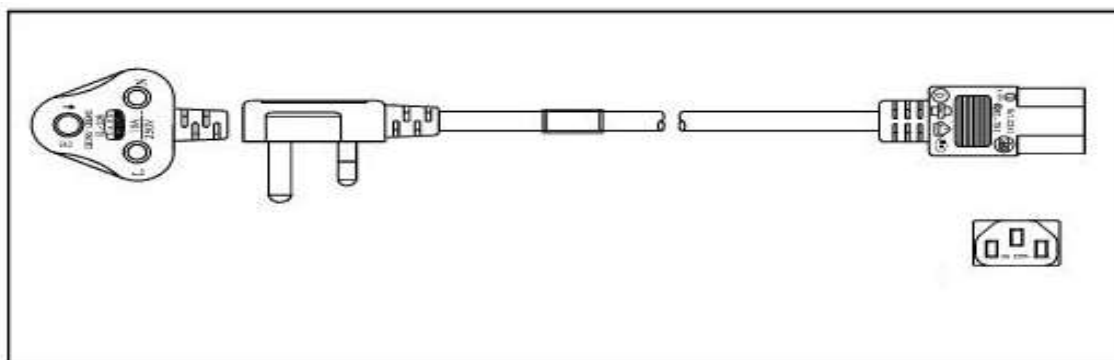


DC-DC Input Power Cable		
SI NO	Pin No.	Connection
1	1,2	VCC
2	3,4	GND

DC-DC Output Power Cable SPECIFICATIONS		
SI NO	Pin No.	Connection
1	1 (RED)	VCC
2	2 (BLACK)	GND

10.2 AC INPUT POWER CABLE

TWSTFT MODEM can be powered ON from AC power Source using the supplied AC power cable.

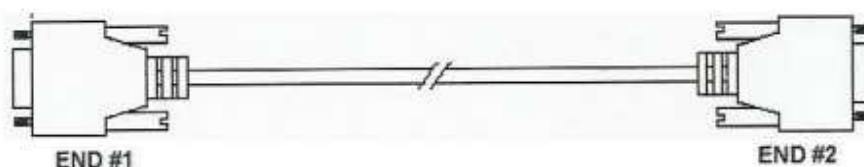


AC Input Power Cable		
SI NO	Pin No.	Connection
1	E	Earth
2	L	Live
3	N	Neutral

AC Output Power Cable		
SI NO	Pin No.	Connection
1	E	Earth
2	L	Live
3	N	Neutral

10.3 RS-422 DATA CABLE

10.3.1 NMEA DATA OUT & 1PPS

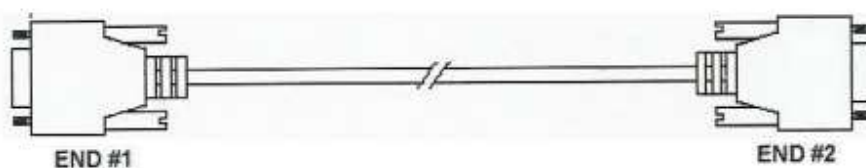


DB-9 Pin No	Name	Type	Description
1	NC	-	Not Connected
2	RS422 TX	Differential Output	NMEA Data OUT Positive polarity
3	RS422 /TX		NMEA Data OUT Negative polarity

SPECIFICATIONS

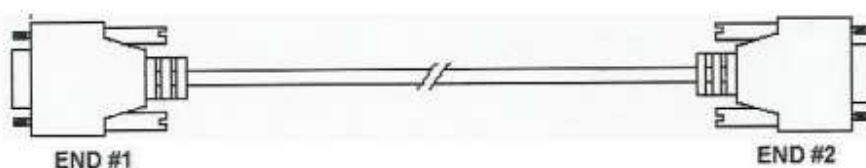
4	NC	-	Not Connected
5	GND	Ground	Device Ground
6	NC	-	Not Connected
7	RS422 1PPS	Differential Output	1PPS Data OUT Positive polarity
8	RS422 /1PPS		1PPS Data OUT Negative polarity
9	NC	-	Not Connected
Shield	Drain	Case Ground	Connected to Chassis Ground

10.3.2 10 MHz OUT



DB-9 Pin No	Name	Type	Description
1	NC	-	Not Connected
2	RS232 TX	Output	Debug for future use
3	RS232 RX	Input	Debug for future use
4	NC	-	Not Connected
5	GND	Ground	Device Ground
6	NC	-	Not Connected
7	NC	-	Not Connected
8	RS422 10MHz	Differential Output	10 MHz Data OUT Positive polarity
9	RS422 /10MHz		10 MHz Data OUT Negative polarity
Shield	Drain	Case Ground	Connected to Chassis Ground

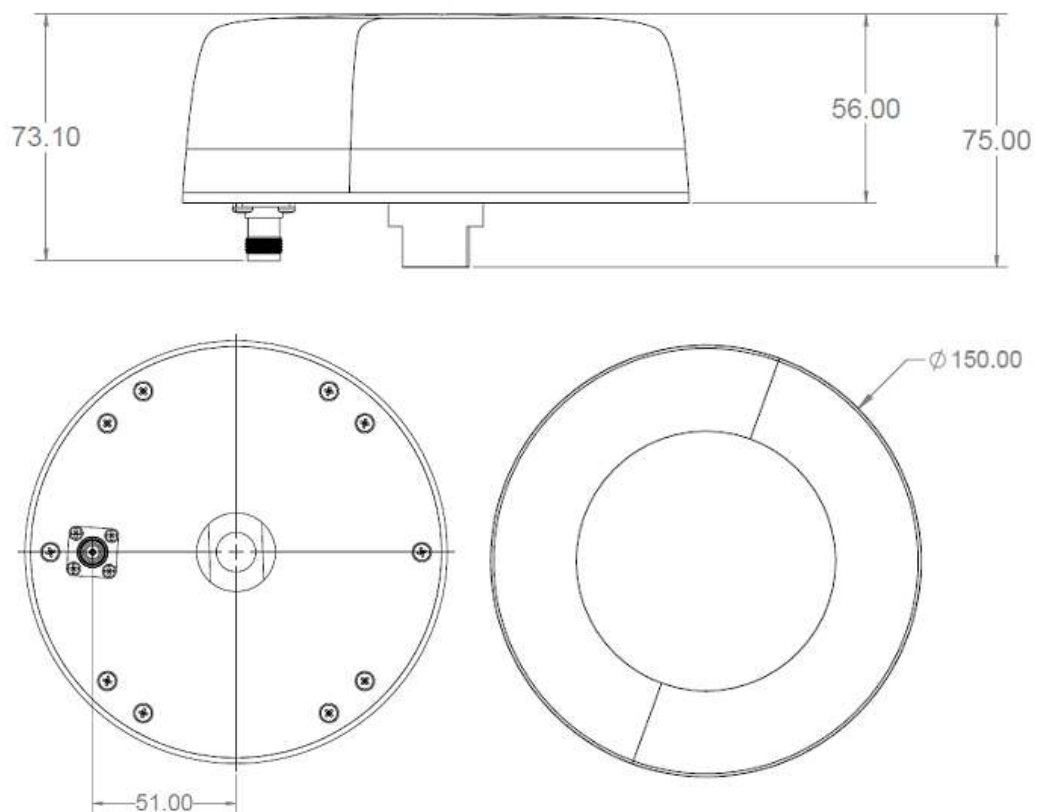
10.3.3 NMEA DATA IN



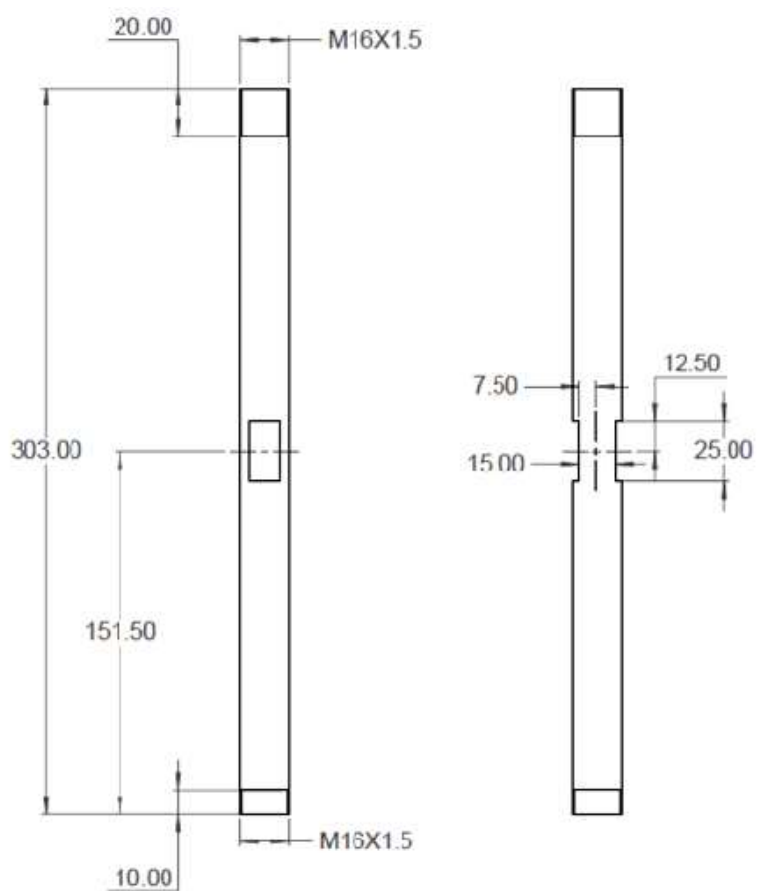
DB-9 Pin No	Name	Type	Description
1	NC	-	Not Connected
2	RS422 RX	Differential Input	NMEA Data IN Positive polarity
3	RS422 /RX		NMEA Data IN Negative polarity
4	NC	-	Not Connected
5	GND	Ground	Device Ground
6	NC	-	Not Connected
7	NC	-	Not Connected
8	NC	-	Not Connected
9	NC	-	Not Connected
Shield	Drain	Case Ground	Connected to Chassis Ground

11 MECHANICAL SPECIFICATIONS

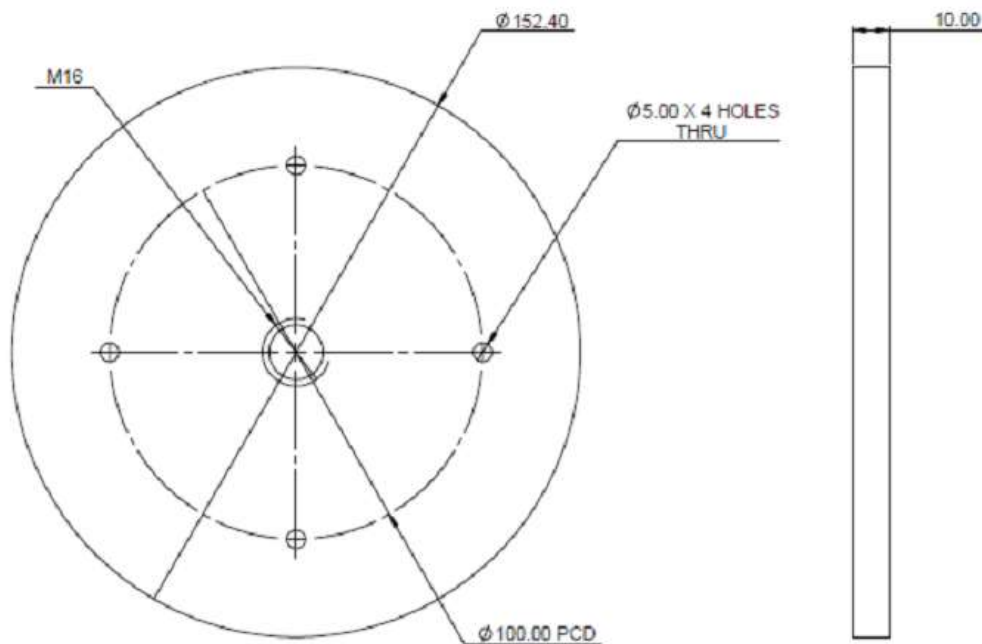
11.1 ANTENNA



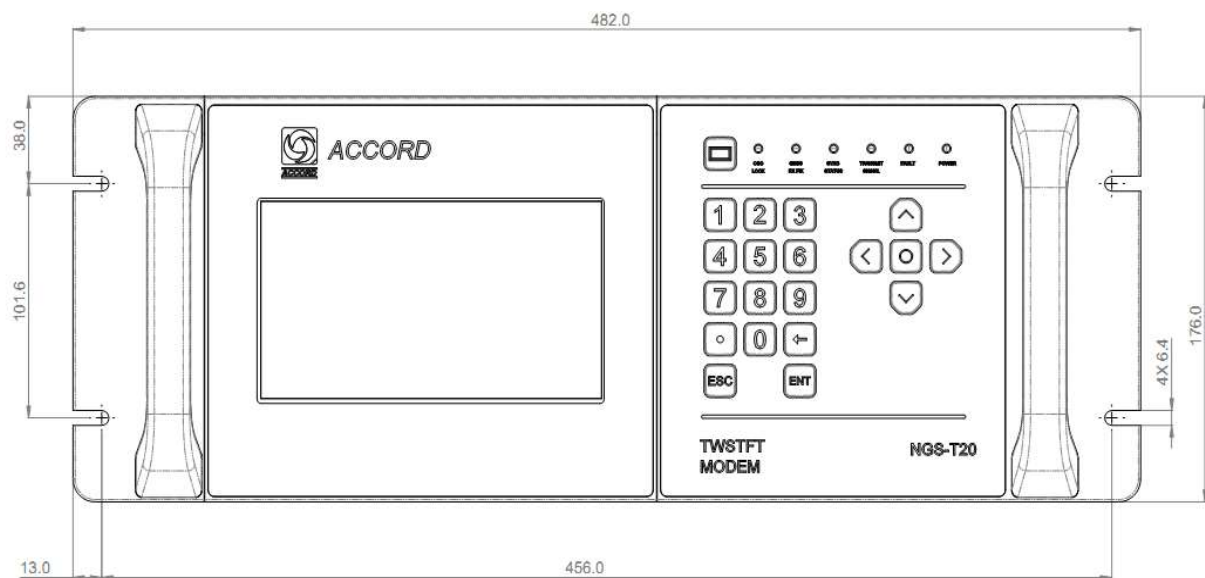
11.2 ANTENNA MOUNT



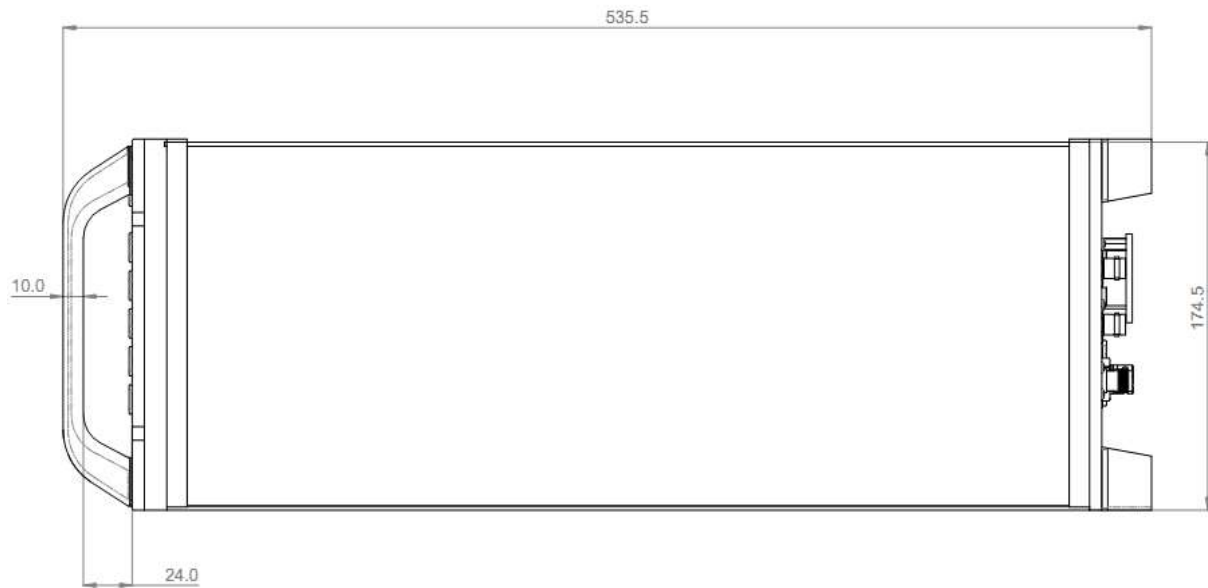
SPECIFICATIONS



11.3 MODEM



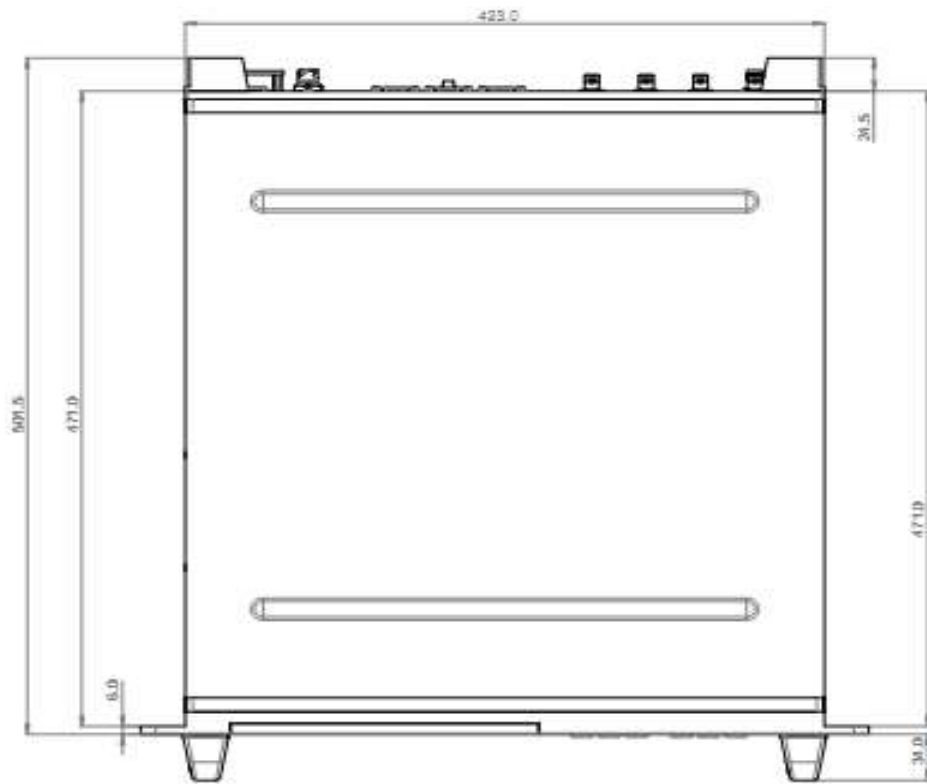
SPECIFICATIONS



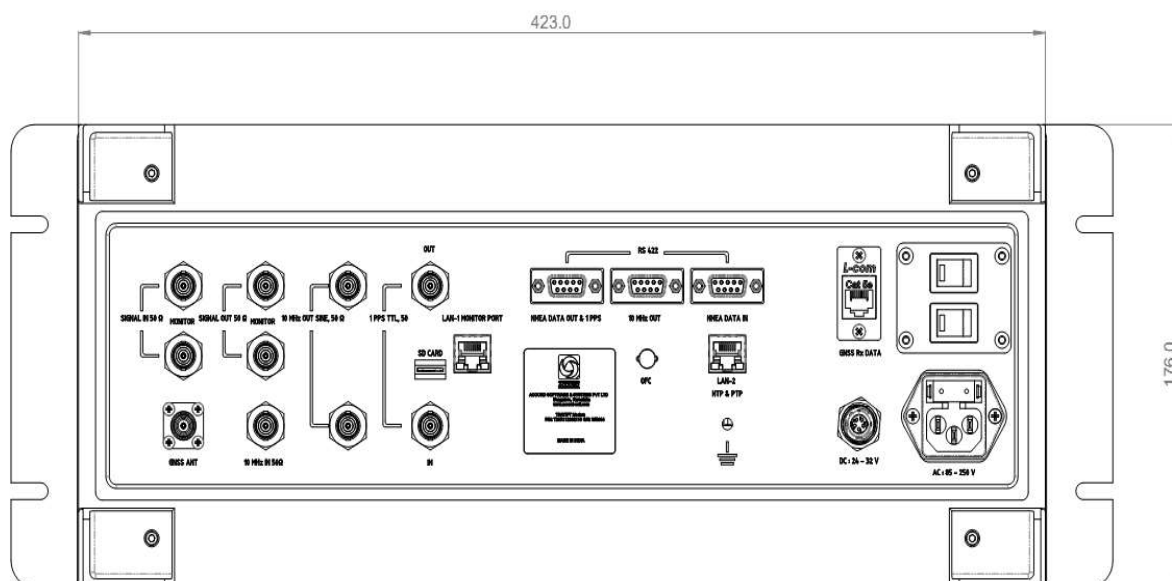


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SPECIFICATIONS



SPECIFICATIONS



APPENDIX:

12 TROUBLESHOOTING

The front panel LEDs and the Web UI provide TWSTFT MODEM status information that can be used to help troubleshoot failures that may occur.

LED	STATUS	IMPLICATION	TROUBLESHOOTING
POWER	LED is OFF	MODEM has no Power AC or DC input.	1. Verify AC power is connected to AC source. And switch is turned ON. 2. Verify DC power is connected to DC source with Specified operating Range. And switch is turned ON.
OSC LOCK	LED is OFF	Oscillator lock is not available.	1. Make sure the Rubidium oscillator is powered ON. 2.
GNSS RX FIX	LED is OFF	GNSS Receiver position fix not available	1. Make sure GNSS Antenna is connected. 2. In DC mode, calculate the power consumed by the MODEM and compare with Specifications Range.
SYNC STATUS	LED is OFF		
TRANSMIT SIGNAL	LED is OFF	No Transmission of IF signal	
FAULT	LED is ON		

TABLE 3: TWSTFT MODEM TROUBLESHOOTING THROUGH FRONT PANEL LED

13 FREQUENTLY ASKED QUESTIONS (FAQ)'S